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A fuzzy multi-criteria framework for AI-driven sustainable technology adoption and economic resilience in pharmaceutical cold chains

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ABSTRACT

The resilience of Pharmaceutical Cold Chains (PCCs) is vital for maintaining the quality and efficacy of temperature-sensitive medications and vaccines. Considering the field of sustainable technology entrepreneurship, the strategic implementation of AI-based innovations within the logistics sector of the pharmaceutical industry is a significant economic choice with clear consequences for organizational sustainability. This study proposes a structured decision-making framework to evaluate and prioritize Artificial Intelligence (AI)-driven technologies through a comprehensive, multi-dimensional approach that enhances PCC resilience across technical, business, environmental, and social dimensions. The framework integrates advanced fuzzy Multi-Criteria Decision-Making (MCDM) methods to systematically assess technologies under uncertainty. This research develops a novel framework for decision-making that makes two main contributions. First, from a methodological perspective, it proposes a *p,q*-Quasirung Orthopair Fuzzy (*p,q*-QOF) extension to the existing Ranking Alternatives with Weights of Criteria (RAWEC) method that incorporates a *p,q*-QOF Delphi-Lawshe's Content Validity Ratio (CVR) to address high-order expert uncertainty and hesitancy. Secondly, from a substantive perspective, it presents a strategic approach to the prioritization of AI-driven technological solutions that identify predictive analytics and ML models as the most impactful solutions for enhancing the resilience of PCCs. This research thus closes the gap between fuzzy modeling theory and resource allocation for high-stakes medical logistics applications. It reveals that while technical robustness is the primary concern for technical systems, sustainability in the multi-dimensional approach to business agility and human capital development is a critical factor for resilience. The findings offer actionable economic guidance for pharmaceutical organizations navigating technology investment decisions under regulatory and operational uncertainty.

Introduction

The cold chain, defined as a temperature-controlled supply network, plays a critical role in safeguarding the quality, safety, and integrity of perishable products (Vrat et al., 2018). Its importance is particularly evident in sectors such as food, pharmaceuticals, and biotechnology, where products are highly sensitive to environmental fluctuations (Aung & Chang, 2014). Under these circumstances, the logistics processes must be carried out while maintaining proper temperature, humidity, and light conditions from the production stage to the distribution stage. Failure to maintain these conditions at any stage of the supply chain would affect the quality and reliability of the product. Cold chain logistics has become the backbone of the supply chain today. Research has proven that the failure to maintain proper temperature conditions would lead to the degradation of the product, resulting in huge monetary losses (Fernández et al., 2017). In the healthcare industry, cold chain logistics is the backbone of the distribution of vaccines and other drugs that have to be stored at proper temperatures. Therefore, the reliability of the cold chain logistics system would have a direct impact on the safety of the public at large (Thakur et al., 2024). Thus, efficient cold chain logistics systems have to be implemented to avoid wastage of the product and to increase the reliability of the supply chain.

However, despite its undeniable importance, the cold chain is also highly prone to disruptions with significant implications. For example,

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even slight changes in temperature can negatively affect the quality and safety of the products (Lorenc et al., 2021). In addition, minor logistical issues (Turan & Ozturkoglu, 2022) can negatively affect the reliability and integrity of the entire supply chain. Other disruptions that can occur in the cold chain include equipment failure, human mistakes, transportation problems, and electricity supply failure. Temperature changes can negatively affect the reliability and efficiency of perishable items (Loisel et al., 2021). For instance, Derens et al. (2006) assert that about 12% of the products during transportation and on the platforms have an average temperature that is higher than the recommended average temperature. During the pandemic period, severe issues were reported across the world concerning the cold chain logistics, especially concerning the vaccines that require ultra-low temperatures (Fahrni et al., 2022). Such issues have negatively affected the reliability and integrity of the cold chains, which are quite sensitive and fragile (Khan and Ali, 2023). The increased interconnection of the world's supply chains is also adding to this vulnerability, which stresses the importance of strategies that can improve the cold chain's resilience and robustness (Khalid et al., 2024).

Artificial Intelligence (AI) strategies have also been identified as being critical and can be transformative with the help of AI, Machine Learning (ML), and other technologies that can improve the capability to anticipate these disruptions (Lau et al., 2021). For example, it can process huge amounts of data in real-time, thus facilitating more precise control of temperature conditions, routing, and decision-making processes (Lorenc et al., 2021). In addition, AI can make it possible to improve predictive maintenance, thus reducing the risks of system failures (Maier et al., 2024). This AI strategy indicates a trend of sustainable entrepreneurship where the organization uses dynamic capabilities to ensure that its technological investment is in line with resilience goals (Lazarte-Aguirre, 2024). The integration of AI in cold chain logistics is not only beneficial in terms of efficiency and reliability but can also make the system more elastic in the event of unforeseen challenges (Golan et al., 2021). In this regard, AI presents itself as a fundamental factor in advancing the development of a more resilient and secure cold chain system.

Pharmaceutical Cold Chains (PCCs) are very sensitive systems since they involve the movement and storage of crucial drugs, such as vaccines and other temperature-controlled drugs, in the healthcare sector. Thus, the integrity of these drugs is paramount, and any change in temperature may make them ineffective or unsafe, and the life of a patient may be at stake (Hulea et al., 2018). AI technology is very crucial in ensuring the resilience of PCCs, especially in the decision-making process, by providing better monitoring, forecasting, and decision-making capabilities (Bamakan et al., 2021). For instance, PCCs can use the Internet of Things (IoT) to monitor the collected data from sensors placed in the system, thereby indicating any abnormalities (Monteleone et al., 2017). Real-time data acquisition facilitates timely intervention strategies such as adjusting temperature settings and shipping routes to prevent food spoilage. Further, the application of AI in predictive analytics can pinpoint disruptions and equipment failure before they happen (Aljohani, 2023). Real-time monitoring can also track transport delays, and administrators can take proactive steps to correct the situation. Big data analytics can also be used to optimize logistics by determining the fastest and most efficient routes and pinpointing what to optimize, which can reduce transit times and the risk of temperature fluctuations (Wang et al., 2016). This improves the PCC, making it even more effective in the face of unexpected disruptions in the supply chain without compromising the safety and efficacy of the product with these technologies.

While the conventional cold chains for other perishable food products are mainly concerned with the extension of product shelf life and prevention of aesthetic degradation, PCCs, on the other hand, are subject to a 'zero-tolerance' environment. Here, temperature-sensitive biologic and vaccine products, such as those needing ultra-low storage during the COVID-19 pandemic, possess complex molecular structures that cannot be altered or reversed, even with small temperature changes. Unlike other logistics, where a failure in the supply chain results in economic loss, a PCC failure directly equates to public health crises, vaccine ineffectiveness, and life-endangering situations for patients.

Conventional technologies fail to handle and manage the entire PCC network due to limited storage and processing capacities and therefore need to evolve to a more advanced level. The emerging technologies are transforming the pharmaceutical industry into modern digital and technology-dependent manufacturing systems (Bellantuono et al., 2021). Drivers of Industry 4.0 can help pharmaceutical firms overcome these challenges, but most firms are still in the early stages of adoption. According to Inamdar et al. (2021), big data analytics is underutilized in the healthcare industry, highlighting the need for further study to address these challenges. Although the technologies provide robust and flexible manufacturing processes, less disruption in medicine production and delivery, and a quick response to drug or supply shortages (Wong et al., 2023), selecting the appropriate technology to meet the requirements has become a critical decision.

Even though there is increasing interest in the optimization of the cold chain, the PCCs face a number of challenges in the decision-making process. PCCs are required to meet the stringent regulations while ensuring the safe distribution of temperature-sensitive biologics, vaccines, and specialty pharmaceuticals. Such factors are associated with a high level of operational uncertainties and risks, and the decisions regarding the adoption of technology are critical. Nevertheless, despite the immense potential of AI-related technologies in addressing issues of cold chain disruption, the pharmaceutical industry still faces a lack of a specialized multi-dimensional approach in determining the relative importance of such capital-intensive decisions, as dictated by the unforgiving nature of medical product regulatory compliance and integrity. The decision-making process is still largely based on the application of technology or the use of conventional low-order fuzzy approaches, which cannot adequately address the high-order hesitancy and associated uncertainties of such decisions in the pharmaceutical industry. This results in a waste of resources, as some technologies may be implemented despite their positive contribution to technical sustainability, with little consideration for essential aspects of social sustainability or environmental sustainability. Hence, there is an urgent need for the development of a universal p,q-Quasirung Orthopair Fuzzy (p,q-QOF)-based approach to decision support, with the aim of ranking AI-related technologies such as predictive analytics and monitoring, ensuring that cold chain architectures are not only technologically advanced but also holistically resilient and operationally viable.

The present study is particularly relevant to an ever-increasing dependence on robust cold chain systems, especially in the pharmaceutical industry, where the stakes are high. As global demand for temperature-sensitive medications and vaccines continues to grow, so does the need for innovative solutions capable of ensuring their safe and effective delivery. Therefore, AI-driven technologies represent a highly promising approach for enhancing the resilience of the cold chain. Despite their potential, there is a notable gap in the literature regarding the systematic evaluation and prioritization of these AI-driven strategies. This research aims to address this gap by providing a comprehensive review of AI techniques applied in Cold Chain Management (CCM) through the lens of p,q-QOF Multi-Criteria Decision-Making (MCDM) methods, offering significant contributions from both academic and practical perspectives. To fully represent the high level of uncertainty and hesitation in expert judgments, this work employs a p,q-QOF model. Compared with traditional fuzzy and intuitionistic fuzzy models, the p,q-QOF model has greater flexibility in representing the membership, non-membership, and hesitation degrees of the fuzzy sets, which is helpful in representing the complexity of the uncertainties in the judgments of the experts on the AI-based technologies in the context of PCC resilience.

Beyond its technical advantages, the p,q-QOF framework can be viewed as a meaningful conceptual improvement in modeling uncertainty and expert hesitation in supply chain resilience decision-making.

In contrast to IFSs and PFSs, where membership and non-membership are typically treated in a more symmetric or fixed-form manner, *p,q*-QOFSs allow confidence and skepticism to be modeled independently through the parameters *p* and *q*. This added flexibility is particularly relevant in PCC contexts, where overestimating and underestimating a technology's contribution to resilience are distinct and not equally treated cognitively. Moreover, by explicitly modeling hesitancy as a separate component rather than treating it as a residual term, the *p,q*-QOF framework offers a more realistic representation of expert judgment under uncertainty, especially when evaluating emerging AI technologies in environments characterized by regulatory and operational ambiguity. Overall, this richer representation of uncertainty is expected to improve the validity of criteria weighting and enhance the robustness and reliability of the resulting technology rankings.

First, a Ranking Alternatives with Weights of Criteria (RAWEC) approach based on the *p,q*-QOF sets (*p,q*-QOFSs) is developed, extending prior models by incorporating expert-based weighting and uncertainty handling. To do this, we adapt the conventional Delphi technique utilizing the *p,q*-QOF operators, and then, employing these operators alongside a score function-based procedure, we determine the weights of the criteria. Finally, the RAWEC model is extended using *p,q*-QOF information to evaluate and rank prospective AI-driven methods. To date, the RAWEC method has been introduced from the perspectives of crisp and fuzzy information, which have not considered the importance of criteria and the weights assigned by experts. The determination of criteria and the assignment of weights to experts are critical components that significantly influence the accuracy of decisions made during the decision-making process. Furthermore, these studies have been unable to address the *p,q*-QOF information.

The research is driven by three mutually connected goals. Firstly, the study aims to define and verify important resilience attributes specific to PCCs using structured expert consultations and fuzzy decision-making methods. In this way, the researcher seeks to ensure the reliability of the criteria set as the basis for the evaluation of the multidisciplinary importance of each factor in terms of pharmaceutical cold chains. Secondly, the study seeks to estimate and rank the efficiency of seven artificial intelligence-based technologies in solving the stated resilience issues. The evaluation process is performed using the *p,q*-QOF MCDM approach as an effective means of accounting for uncertainty in experts' assessment of new technologies. Thirdly, the research produces evidence-based insights for making managerial decisions in the context of technology-intensive supply chain management.

The key contribution of this research does not lie in its use of a more advanced mathematical fuzzy logic approach; rather, it lies in the fact that such an approach results in superior decisions for those involved in managing the cold chain logistics of the pharmaceutical industry. There are three advantages of the current approach over the previous one that can be highlighted. Firstly, the independent calibration of the sensitivity factors p and q of membership and non-membership, respectively, ensures that technologies having entirely different risk characteristics are not treated equally, something that would have occurred in the case of symmetric fuzzy logic. The significance of this difference is not merely theoretical to the extent that a manager investing resources in competing uses of AI needs to make a choice with respect to which of these implementations will involve risks, either evident at the time of the decision-making process or concealed from the decision-maker. Secondly, by embedding the CVR and Delphi methods of validation into the evaluation process, the current research ensures that only those criteria that are shown to be essential according to the expert judgment of scholars would determine the prioritization. The third advantage, which follows from sensitivity analysis, is the stability of prioritization and therefore its usefulness as an investment criterion, considering that technological decisions are resource-intense and not easy to change.

It is important to note that the purpose of this study is exploratory and strategic rather than confirmatory. The study does not aim to provide large-sample empirical validation of AI technology performance outcomes, nor does it seek to generalize the ranking results to the global pharmaceutical cold chain context. Instead, it is designed to support expert-driven validation and to generate theoretically informed prioritization insights for decision-makers operating under high uncertainty, particularly where operational data on emerging technologies are limited and regulatory constraints are significant. Within this methodological framing, rigor is defined differently from large-sample empirical studies. It is reflected in the validity of the expert consensus process, the analytical robustness of the fuzzy decision-making framework, and the stability of results under sensitivity analysis. Accordingly, this scope is not a limitation but a deliberate design choice, aligned with established approaches in strategic decision-making research based on expert judgment (Thakur et al., 2024; Lau et al., 2021).

The subsequent sections are structured as follows. Section 2 reviews prior studies relevant to this research. Section 3 outlines the fundamental concepts taken into account in this study. Additionally, the *p,q*-QOF properties as well as the extension of the RAWEC method are described. Section 4 presents the results, while Section 5 discusses the implications of the study. Section 6 presents the conclusions of the study and offers recommendations for future research endeavors.

Literature review

Pharmaceutical cold chain resilience: dimensions and theoretical foundations

Advanced technologies can significantly enhance cold chain resilience in the pharmaceutical industry. By leveraging digital tools and techniques, companies can better monitor risks, optimize processes, and maintain the integrity of temperature-sensitive products. Joshi et al. (2012) identified the key performance indicators and decision factors for cold chain effectiveness analysis through a methodology based on dual-graph theory for continuous improvement. Ashok et al. (2017) reported on the Clinton Health Access Initiative, Inc. experience with national immunization programs across ten countries to strengthen vaccine cold chains through root cause analysis and resolution of common challenges. They emphasized the use of temperature monitoring and control devices to minimize excursions and equipment breakdowns.

Supply chain resilience is commonly understood as the ability of the chain to anticipate, defend against, absorb, and recover from shocks while ensuring continuity of essential operations (Tukamuhabwa et al., 2015). In the case of PCCs, this concept acquires greater relevance since, unlike typical supply chains, PCCs must comply with zero tolerance where disruption compromises the effectiveness of the product and the well-being of patients. Extending the resilience construct proposed for supply chains by Ivanov et al. (2019), this study defines PCC resilience along four integrated dimensions.

The technical dimension involves the functionalities that allow for real-time monitoring, predictive maintenance, and anomaly detection, which are the underlying processes by which disruptions are spotted and controlled. The business dimension concerns agility and adaptability, along with efficiency, the organizational characteristics that help determine the speed with which the PCC can adjust to disruption. The environmental dimension involves sustainable considerations relating to energy usage and the carbon footprint, which have become increasingly significant both from the perspective of regulation and sustained success in pharmaceutical logistics (Arya et al., 2024). The social dimension refers to human resources, workforce preparation, and stakeholder involvement in the organizational context within which technical capabilities must be realized (Raman et al., 2025).

The four dimensions do not represent parallel evaluation dimensions but are interrelated elements forming the resilient architecture system. Technical strength without business agility creates inflexible systems; business agility without social readiness results in untapped potential. The holistic approach justifies the rationale behind the multi-dimensional criterion used in this study and makes it different from previous models that use the one-dimensional framework to measure resilience.

The global development of a cold chain requires coordination between countries with varying incomes, enforcement of policy and regulation, and prevention of beggar-thy-neighbor tactics (Peters & Sayin, 2023). Khan and Ali (2023) identified crisis simulation, logistics identification and security, and digitalization as top strategies for ensuring cold supply chain resilience during COVID-19 disruptions using fuzzy quality function development. Furthermore, some authors have focused on specific products in literature reviews. While some scholars review the cold chain from a broader perspective, others narrow down the literature review to specific products. For instance, Ren et al. (2022) examined contemporary studies on meat packaging, meat quality assessment, and forecasting within cold chain logistics. They spotlighted the significance of optimizing packing materials and dynamic quality perception to attain meat quality and safety. The analysis asserted that comprehensive packaging and astute quality assessment are essential for converting conventional cold chain logistics into smart, sustainable, and efficient cold chain logistics. Tirkolaee et al. (2023) offered a robust decision-support framework integrating a hybrid MCDM approach based on the Best-Worst Method (BWM), Weighted Aggregated Sum Product Assessment (WASPAS), and Type-2 Neutrosophic Fuzzy Numbers (T2NNs) for resilient supplier selection in a pharmaceutical supply chain. They addressed supply chain resilience by incorporating resiliency scores, facility reliability, and uncertainty considerations. Thakur et al. (2024) explored factors influencing vaccination CCM for sustainable healthcare provision during health crises. Employing fuzzy Delphi, fuzzy Analytical Hierarchy Process (AHP), and K-means clustering, they identified cold chain infrastructure, vaccine storage protocols, and qualified personnel as essential components for efficient management. Chen et al. (2022) analyzed contemporary studies on food cold chain logistics and emphasized micro, meso, and macro levels. They highlighted the issues and deficiencies in microenvironmental monitoring, life cycle evaluation, and global effects. It was revealed that emerging trends encompass intelligent, methodical, low-carbon logistics and advanced information technology.

AI-driven technologies for PCC resilience

AI technology facilitates resilience through better prediction capabilities, enhanced operational transparency, faster decision-making capabilities, and faster disruption recovery. The use of AI technology is no longer just an operational tool but a strategic lever for PCC management.

In the context of IoT technologies, Tsang et al. (2018) presented an IoT-based risk monitoring system designed to manage product quality and occupational safety concerns in cold chains, employing Wireless Sensor Networks (WSNs), cloud database services, and a fuzzy logic methodology. The suggested system reduces product quality and occupational safety hazards, enhances ergonomics, and facilitates real-time data interchange in the IoT context, thereby enabling customized and proactive risk management for cold chain stakeholders. Acknowledging the importance of traceability and safety during the pandemic, Qian et al. (2022) established a system utilizing IoT and blockchain technology to enhance food CCM amid the COVID-19 pandemic by forecasting SARS-CoV-2 transmission risk and recording essential control point data for reliable traceability. Jackson et al. (2025) explored the usage of AI/ML forecasting methods in the cold chain capacity planning domain. In the cold chain domain, the study focused on temperature control. Lorenc et al. (2021) explored the temperature profile and determined the gradients as well as the causes. A mathematical model was proposed for automatic identification through the usage of an artificial neural network for complex transport curve determination. Lau et al. (2021) proposed a Federated Learning-enabled Multi-criteria Risk Evaluation System (FMRES) for the evaluation of risks in cold supply chains. It was developed by integrating the fuzzy logic approach with BWM to determine the risks of individual entities through the hierarchical structure of the risk framework. It comprehensively included factors like technology, equipment, operations, external environment, and personnel in the evaluation process. Furthermore, Kim et al. (2016) proposed an intelligent framework for the management of risks in cold chain logistics, named i-RM. It was designed for real-time and context-aware management of risks to cover different scenarios of risk occurrences. In the context of emerging economies, Karmaker et al. (2023) applied a fuzzy model based on Total Interpretive Structural Modelling (TISM) and Cross-Impact Matrix Multiplication Applied to Classification (MICMAC) to identify and analyze supply chain risk factors. Their findings showed that factors such as management commitment to sustainability and workforce capabilities play a highly influential structural role in shaping overall supply chain risk. This is consistent with the present study, where human capital is similarly assigned a high level of importance within the prioritization results.

Zhang and Mohammad (2024) conducted a study to ascertain the influence of sustainability innovation and supply chain resiliency on the economic, environmental, and social performance of Chinese cold chain logistics organizations. The results of the study revealed that sustainability, innovation, and supply chain resiliency are beneficial for economic, environmental, and social performance. Aladaileh and Lahuerta-Otero (2025) consider issues such as demand fluctuations and price competition by employing multi-criteria decision analysis techniques to evaluate risk in sustainable supply chain innovation within the Jordanian garment, textile, and leather industries. Ochieng et al. (2024) conducted a study to determine the capability of Generative Artificial Intelligence (GAI) and Machine Learning (ML) algorithms in increasing productivity in the oil and gas industry's value chain. Using Principal Component Analysis and Structural Equation Modeling, the authors of the study proved the effectiveness of GAI and ML algorithms in the upstream, midstream, and downstream sectors of the oil and gas industry's value chain. Recently, Jafarian et al. (2025) proposed a data-driven model for designing a sustainable, resilient, and digitalized pharmaceutical supply chain using ML and a multi-objective optimization model. Key indicators like delivery time, quality, and cost were prioritized, and a blockchain-based information-sharing system was selected in a real-world case study in Iran. These various studies highlight the convergence of advanced technologies, risk management, and collaborative strategies, particularly the integration of IoT, AI, ML, and sustainability, as a unifying trend driving global advancements in cold chain resilience and efficiency.

Throughout the existing literature in the field of supply chains, the potential of AI technologies for enhancing resilience through various functional dimensions has been studied: anticipation, absorption, and recovery. While such analysis is important, previous studies have not provided an integrated approach to evaluating these technologies for enhancing resilience, focusing only on one functional dimension and using different criteria for assessing the performance of each AI technology. In other words, previous works lack an effective framework that would allow researchers to assess the relative performance of different AI technologies from the perspective of enhancing resilience based on multiple dimensions at once. Such a framework is crucial because, although some technologies might perform very well according to technical robustness criteria, they can be inefficient in terms of social readiness or environmental sustainability.

Technology prioritization and decision-making challenge

Although more AI-enabled technologies have been developed, identifying the best solutions to adopt in PCC settings represents an extremely challenging strategic decision problem that includes many conflicting considerations. Nonetheless, the current literature has assessed the efficacy of these techniques separately without providing a common platform for assessing and contrasting their respective efficacy across all four resilience dimensions. The significance of having such a platform is vital because a technology that may be considered highly effective in terms of technical criteria may prove to be ineffective in other aspects. Therefore, this paper attempts to fill this important gap in the existing literature by integrating the assessment of all seven AI technologies into one MCDM framework addressing all four resilience dimensions.

Seikh and Mandal (2022) developed the *p,q*-QOFSs, an extension of the q-rung orthopair fuzzy sets (q-ROFSs). Subsequently, various researchers suggested q-ROFSs-based aggregation procedures using operators such as the Aczel-Alsina (AA) operation (Ali & Naeem, 2023), generalized Dombi aggregation operators (Ali & Mehmood, 2025), sine-trigonometric operational laws (Rahim et al., 2023), cosine similarity and distance measures (Rahim et al., 2024a), and *p,q*-cubic QOF operators (Chu et al., 2024). Additionally, Rahim et al. (2024b) offered a novel expansion of the COPRAS approach employing *p,q*-QOFSs. You et al. (2024) investigated novel non-linear programming models employing the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method under cubic *p,q*-QOFSs for green supplier selection. Recent reviews also highlight the rapid expansion of q-ROFS-based MCDM methods in handling complex decision problems (Öztaş, 2025).

The third research aim in this study is based on strategic decision-making literature and technology adoption literature that have a common concern in terms of how organizations deploy their limited resources among competing technology options in uncertain situations. The classical theory of decision-making holds that individuals make choices to maximize expected utility based on available information. However, technology adoption decisions in regulated and capital-intensive industries like pharmaceutical logistics are inherently subject to bounded rationality, information asymmetry, and conflicting stakeholder interests, which cannot be captured by classical approaches (Lazarte-Aguirre, 2024).

The problem of resource allocation in such cases is much more complex than simply financing the technologies, as it requires considering the availability of organizational resources and capacity for staff development and infrastructure preparation, which are limited and in direct competition with operational considerations. Existing literature on technology implementation failure suggests that it has been more common for the issue to arise from incorrect resource allocation than technology itself; specifically, allocating technological innovations within an organization at the incorrect stage of its development has proven to be the main reason for the problem (Samuels, 2025). This consideration prompts the third research objective and makes it necessary to develop a context-specific resource allocation-based guideline for implementing AI technologies.

Taken as a whole, the literature review above highlights several recurring themes. First, the current models make use of additional mathematical space for expert judgment representation regardless of the epistemic premises underlying such judgments' formation within the context of the highly regulated environment. Second, the lower-order fuzzy logic models are based on symmetric epistemic premises; third, the residual hesitancy approach amalgamates different positions into one; fourth, literature-driven criteria sets lack validation procedures; fifth, evaluations of technology tend to follow step-by-step, dimension-by-dimension procedures instead of systemic, holistic evaluations. The novel *p,q*-QOF RAWEC framework suggested in this work aims at addressing all these issues in the context of PCC technologies evaluation. Not through increasing levels of complexity, but through making sure that the epistemic premises behind judgments are addressed and incorporated correctly in the decision-making process.

Research gaps

Several research gaps exist in the current literature regarding the integration of AI in enhancing the resilience of PCCs. Our review of the literature shows that while some AI techniques have been applied to aspects of the cold chain, their adoption within comprehensive resilience frameworks remains limited. The existing applications can be broadly categorized as follows:

- **Forecasting and Predictive Models:** Several works have utilized specific ML models for predictions, such as using Prophet and Seasonal Auto-Regressive Integrated Moving Average (SARIMA) for capacity planning or artificial neural networks for temperature deviation detection and modeling.
- **Risk Assessment Systems:** Researchers have proposed intelligent systems for risk assessment by applying fuzzy logic in conjunction with IoT data or federated learning to assess risks hierarchically.

However, there remains a significant gap in the literature, as these AI technologies are often applied in isolation to address specific problems such as capacity planning or temperature monitoring. There is limited research on integrated frameworks that combine different AI domains, including predictive machine learning, anomaly detection, and optimization techniques, within an integrated structure to address the complexity of PCC resilience. In addition, existing studies do not sufficiently address the need for multi-criteria evaluation approaches that can assess these technologies collectively within a strategic decision-making context. This study addresses this gap by proposing an integrated framework that enables the systematic evaluation and prioritization of multiple AI technologies for PCC resilience.

There is a lack of multi-criteria evaluations, which consider technical, business, environmental, and social criteria simultaneously while assessing the impact of AI-driven technologies on PCC. In this regard, the current literature fails to take into account the multi-dimensional aspects of AI-driven evaluations, which are essential to understand the impact of different strategies on cold chain resilience. To fill this literature gap, the current study aims to develop a hierarchical structure of criteria and strategies, which is critical to understanding the interrelationship between different criteria in complex decision scenarios. The lack of such a structure impedes the formulation of effective guidelines for technology prioritization. Moreover, environmental and social aspects are also found to be underrepresented in cold chain studies, with most studies concentrating on technical and business aspects to the exclusion of sustainability and social issues, which are becoming increasingly vital in contemporary supply chain management (Priyan & Udayakumar, 2025). This phenomenon is in line with the broader trends in the technology sustainability literature, in which the dimensions of environmental and social sustainability are seen as secondary considerations in decision frameworks compared with technical and economic criteria (Arya et al., 2024). Additionally, the special features of PCCs, such as regulatory requirements and product integrity, are also found to be insufficiently addressed in cold chain studies, leading to results that may not be fully satisfactory to the pharmaceutical industry.

In addition, few studies have emphasized AI-related technologies based on their individual contribution to the PCC. In fact, the majority of existing studies did not offer a ranking based on the importance of each technology for the industry, which is a limitation for decision-makers in the field. Yet another limitation is the lack of use of more advanced methods of fuzzy MCDM. Although the use of fuzzy MCDM techniques has been successful in many fields (Božanić et al., 2025; Gül et al., 2025), their application in prioritizing AI-based strategies for improving the resilience of the cold chain is limited. Particularly, the integration of the *p,q*-QOF Delphi method with Lawshe's Content Validity Ratio (CVR) test is rare, making the precision and reliability of expert judgments in complex decision-making situations limited. Lastly, an extension of the RAWEC method using *p,q*-QOFSs is presented, based on evaluations by experts. Many theoretical frameworks do not sufficiently engage experts in their development, making them less applicable in real life.

A critical analysis of this literature identifies five interrelated gaps that the current research aims to address. First, fuzzy MCDM models used for evaluating supply chain technologies depend mostly on low-order fuzzy sets, which create symmetric restrictions on expert opinions, thus limiting their ability to incorporate asymmetry into the PCC technology evaluation framework. In addition, hesitance is usually considered the remainder of a mathematical operation and not a valid epistemic condition that might influence prioritization outcomes independently. Methodologically, few researchers include expert opinion verification, such as checking for agreement among experts in the group, as one of the mandatory phases of the evaluation process. Substantively, no existing research focuses on systematically comparing multiple AI-driven technologies using all four resilience dimensions, namely technical, business, environmental, and social, as part of a unified evaluation framework. Finally, the practical applicability of MCDM models is often overlooked, as most scholars provide rankings but lack the specific recommendations that managers need to translate prioritization outcomes into feasible implementation strategies.

Current applications of fuzzy in supply chains exhibit three key limitations that are addressed in this framework. First, lower-order fuzzy models impose symmetric constraints on how expert judgments are represented, which can obscure important differences in epistemic positions. Second, treating hesitancy as a residual component rather than an explicit form of judgment can introduce bias in criterion weighting, limiting the ability to fully capture expert assessments of emerging and operationally immature technologies. Third, many existing approaches lack a formal mechanism for validating the criteria used in the weighting process, raising concerns about whether selected criteria truly reflect expert consensus on what is important. This study addresses these limitations in a way that goes beyond increasing methodological complexity. Specifically, the independent calibration of p and q allows asymmetries in implementation risk to be explicitly captured, which may otherwise be hidden in symmetric models. In addition, the Delphi-CVR filtering process ensures that only criteria agreed upon by experts as essential are included in the evaluation. Finally, sensitivity analysis demonstrates that rank stability provides evidence of robustness, which supports more reliable decision-making in high-stakes, capital-intensive investment contexts.

From a conceptual perspective, the progression from deterministic methods such as AHP and TOPSIS, through classical fuzzy sets, IFSs, and PFSs, to q-ROFS reflects an increase in representational flexibility. However, this progression does not fully resolve the underlying conceptual challenge. Each generation of decision models relies on specific assumptions about how expert judgment can be represented. Deterministic approaches assume that preferences can be expressed precisely. Classical fuzzy sets treat uncertainty as one-dimensional. IFSs and PFSs extend this by modeling membership and non-membership simultaneously but still rely on a largely symmetric structure of uncertainty representation. None of these assumptions fully captures the conditions present in PCC technology evaluation, where experts must simultaneously deal with regulatory uncertainty, limited operational familiarity, and asymmetric perceptions of risk. In this context, the p,q-QOF model offers an improvement not simply because it is more general but because it aligns more closely with the actual structure of expert reasoning under these conditions.

This study directly addresses these critical gaps by proposing a comprehensive framework that efficiently integrates multi-dimensional criteria, advanced fuzzy MCDM methods, and structured expert consensus, specifically created for PCC logistics. While many decision-making techniques have been employed in the evaluation of technology in the supply chain, certain limitations have been identified when these techniques are used in the PCC environment. For example, techniques like AHP and TOPSIS are based on deterministic principles and may not be effective in dealing with the uncertainties associated with expert opinions in the evaluation of new technology. Classical fuzzy sets have attempted to address the limitations of these techniques by introducing the concept of membership degrees; however, they may not be effective in dealing with the hesitancy of expert opinions. Even more advanced techniques like q-ROFSs are based on single-parameter structures and may not be effective in dealing with the uncertainties associated with expert opinions in the evaluation of PCC technology.

The lack of systematic frameworks for prioritization of AI-based technologies in the PCCs could have considerable practical consequences:

- **Failure to Model Complex Hesitancy:** Most fuzzy MCDM approaches are based on lower-order mathematical structures that cannot model the unprecedented levels of uncertainty and professional hesitancy observed in the field of medical logistics.
- **Dimensional Imbalance:** Most models are based on either purely technical or economic efficiency criteria, without considering the social sustainability and environmental criteria, which are now critical in the context of modern resilient supply chains.
- **Lack of Empirical Validation:** Much of the theoretical work on MCDM has not sufficiently engaged with multidisciplinary expert panels, and this has limited the practical applicability of the proposed ranking systems.

Motivations of the work

The motivation behind the research is based on the increased challenges that the pharmaceutical industry has been trying to grapple with in the maintenance of resilient cold chains. As the world's demand for temperature-sensitive products such as vaccines, biologics, and specialty medicines rises, more is required to preserve the integrity of these products from the point of production to delivery. The recent disruptions, as a result of logistical complexity, technological constraints, and unforeseen global events relating to the COVID-19 pandemic, have surfaced with significant vulnerabilities in the cold chain infrastructures that lack the required flexibility and robustness to prevent the degradation of the products, which can lead to serious public health and economic consequences.

With the current technological advancements in this area, AI technologies are likely to improve their capabilities in prediction, which will be helpful in enhancing operations through real-time insight (Rame et al., 2024). This, therefore, shows great potential in enhancing the resilience of the cold chain. However, interestingly, there are only a few works of literature that discuss the evaluation of the effectiveness of the technologies in the cold chain context. Most of the literature focuses on general improvements of the supply chain and/or technological solutions, but not all the various aspects of resilience are considered.

Our motivation for choosing this particular topic is based on the need to bridge the identified gap through the integration of AI strategies and enhanced decision-making methodologies. Through the application of fuzzy MCDM approaches, it is possible to develop a critical understanding of the most impactful AI-based solutions in support of strengthening the cold chain. This research is significant in addressing the pressing needs of industry and contributing to the academic body of knowledge in providing a structured approach for evaluating and prioritizing technology interventions. The objective is to provide valuable insights for policymakers, industry leaders, and scholars in strengthening the cold chain for the efficient delivery of critical pharmaceutical products around the world.

This work utilized a novel p,q-QOFSs MCDM approach to enhance and extend traditional fuzzy MCDM techniques in the following ways:

I. **Higher Accuracy in Handling Expert Uncertainty:** Unlike fuzzy sets, the concept of p,q-QOFS provides a more flexible way of handling expert opinions through the concurrent use of membership, non-membership, and hesitancy degrees. This results in the effective handling of ambiguity and uncertainty in decision-making situations.

II. **Integration of Robust Expert Validation Techniques:** The integration of the p,q-QOF Delphi method and Lawshe's CVR is unique in the proposed approach and has resulted in the remarkable reliability and objectivity of expert opinions. This validation method is effective in the selection of the most critical and relevant criteria.

III. **Ranking Consistency:** Through the RAWEC method in the *p,q*-QOF approach, it is possible to enhance the consistency and robustness of alternative rankings under various scenarios of weight and expert opinions. Comparative and sensitivity analyses prove the robustness and reliability of the developed model in comparison to other fuzzy decision-making methods.

IV. **Empirical and Practical Applicability:** By combining expert validation with advanced fuzzy decision-making approaches, this framework provides actionable insights specifically designed for PCCs.

Equally important is the theoretical logic that integrates the four resilience dimensions used in this study. Rather than treating technical, business, environmental, and social criteria as separate and parallel categories, the framework conceptualizes them as interdependent elements of a broader resilience system (Tukamuhabwa et al., 2015; Ivanov et al., 2019). In this structure, technical robustness provides the operational foundation, business agility supports adaptive response, environmental sustainability ensures long-term viability, and social readiness determines whether technological capabilities can be effectively implemented in practice. This interdependence suggests that focusing on any single dimension in isolation may lead to an overestimation of a technology's true contribution to resilience. To address this, the *p,q*-QOF RAWEC framework evaluates and prioritizes AI technologies across all four dimensions simultaneously within a unified fuzzy decision-making structure. In doing so, resilience is treated as an emergent system-level property rather than a simple aggregation of individual scores.

In the absence of a prioritization framework, the pharmaceutical companies may end up investing in the wrong technology, which may not address the key resilience factors, resulting in substantial economic losses. Moreover, the absence of an integrated decision system results in the reactive rather than the proactive management of the supply chain, making the chain vulnerable to equipment failures and power outages, which may compromise patient safety. In addition, the absence of the evaluation process in terms of the "Adopters' acceptance rate" results in the implementation of the wrong technology, which the workforce may end up rejecting, hence stalling the digital transformation process. Inefficient resilience planning in PCCs results in temperature excursions that may violate stringent regulatory requirements, resulting in the complete loss of high-value vaccine batches.

Research methodology

This section outlines the proposed MCDM framework comprehensively. This study briefly discusses *p,q*-QOFSs and demonstrates the basic algorithm of the p,q-quasirung RAWEC method, which is the proposed model. The use of an expert-based evaluation approach is particularly relevant in the context of emerging technologies and strategic supply chain decision-making. In PCC supply chains, several AI technologies are at various stages of maturity and adoption, and there may not be sufficient data on their long-term operation. Therefore, expert opinions can form an important basis for evaluating the potential contribution of these technologies to supply chain resilience. The conceptual framework is illustrated in Fig. 1. Accordingly, the framework operates as a multi-stage 'audit trail' for strategic technology prioritization.

The framework is comprised of three critical phases:

Phase I: Identification and Validation: In this phase, a $p,\ q$-QOF Delphi method is employed to filter the initial set of 47 criteria. This phase guarantees that the selection is informed by multi-dimensional factors such as technical, business, environmental, and social resilience.

Phase II: Essentiality Filtering: To address the issue of expert disagreement for 'moderate' criteria, Lawshe's CVR is employed. This phase acts as a statistical gatekeeper to guarantee that only 'essential' factors are considered for the final selection.

Lawshe's CVR was selected as the criterion filtering method for three main reasons. First, entropy-based approaches, such as Shannon entropy, are designed to derive criterion weights from the variability of performance scores in the decision matrix. While effective for weighting, they are not intended for validating whether a criterion should be included in the first place. Second, consensus measures like Kendall's W assess the level of agreement among experts but do not provide a clear, criterion-level decision rule for inclusion based on a statistical threshold. As a result, they cannot resolve cases where a criterion is viewed as only moderately important in a transparent and consistent way. Third, CVR is specifically developed for use with expert panels to evaluate the essentiality of criteria, which aligns directly with the objective of this phase of the framework. For a panel of 14 experts, the CVR threshold of 0.51 ($p < 0.05$) offers a well-established and rigorous basis for inclusion, ensuring that only those criteria considered essential by the experts are carried forward into the RAWEC prioritization stage.

Phase III: Prioritization via $p,\ q$-quasirung RAWEC: In the final

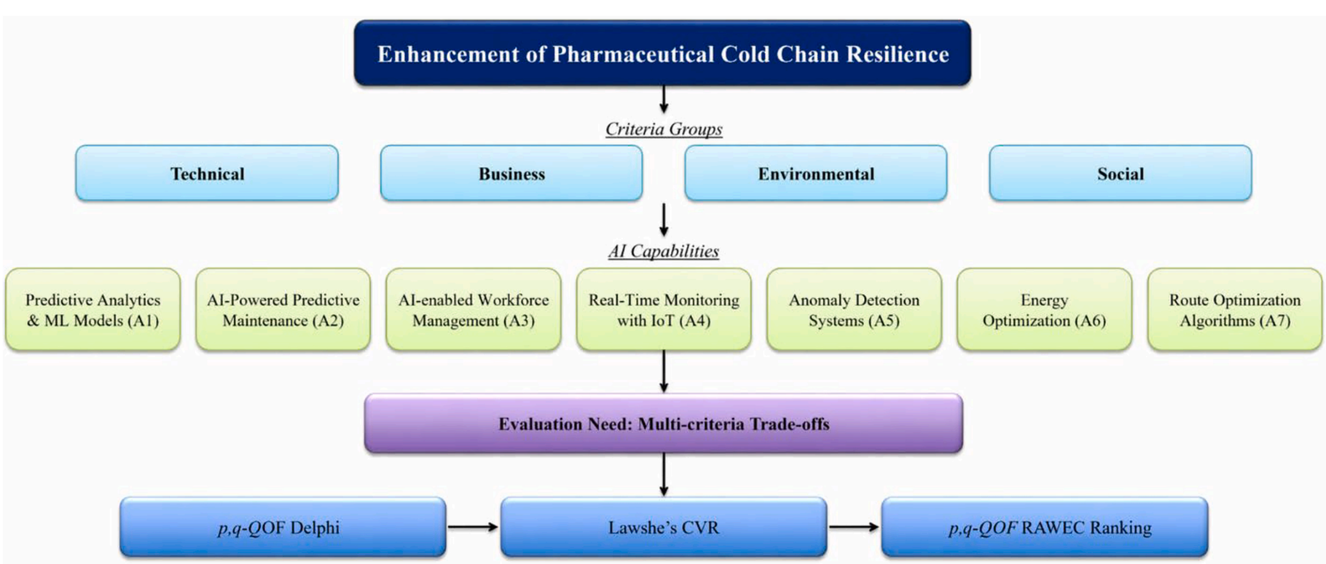


Fig. 1. Conceptual framework for evaluating AI-driven technologies for PCC resilience.

phase, the RAWEC model is extended to employ p, q operators to rank the seven AI strategies. This phase combines the weights from the experts with the scores to identify the most promising technologies for enhancing overall resilience.

Fundamentals of p,q-QOFSs

This section covers a description of the definitions and operational rules of *p,q*-QOFSs essential to our proposed MCDM model (Seikh & Mandal, 2022).

Definition 1. Let T be a non-empty set. A p,q-QOFS$\mathfrak{I}$over an element t is defined as Eq. (1):

$$\mathfrak{I} = \{t, \langle a_{\mathfrak{I}}(t), b_{\mathfrak{I}}(t)\rangle | t \in T\}. \tag{1}$$

The membership and non-membership grades are denoted by $a_{\mathfrak{I}}(t)$ and $b_{\mathfrak{I}}(t)$ of p,q-QOFSs in Eq. (1), where $(a_{\mathfrak{I}}(t))^p + (b_{\mathfrak{I}}(t))^q \le 1$ for every element and $p,\ q \ge 1$. The hesitancy degree is denoted by $\pi_{\mathfrak{I}}(t) = \sqrt[k]{1 - (a_{\mathfrak{I}}(t))^p - (b_{\mathfrak{I}}(t))^q}$. For the sake of simplicity, $(a_{\mathfrak{I}}(t),\ b_{\mathfrak{I}}(t))$is referred to as a p,q-QOF Number (p,q-QOFN).

The following operational explanations explain the procedure and functional connections between p,q-QOFNs:

Definition 2. Let $\eta_1 = (a_1, b_1)$and $\eta_2 = (a_2, b_2)$be the two p,q-QOFNs, then some operations on p,q-QOFNs are defined as

(i) $\eta_1 \oplus \eta_2 = \left(\sqrt[p]{a_1^p + a_2^p - a_1^p a_2^p}, b_1 b_2\right)$,

(ii) $\eta_1 \otimes \eta_2 = \left(a_1 a_2, \sqrt[q]{b_1^q + b_2^q - b_1^q b_2^q}\right)$,

(iii) $c b_1 = \left(\sqrt[p]{1 - (1 - a_1^p)^c}, b_1^c\right)$, where $c \ge 0 \in \mathbb{R}$,

(iv) $b_1{}^c = \left(a_1^c, \sqrt[q]{1 - (1 - b_1^q)^c}\right)$, where $c \ge 0 \in \mathbb{R}$.

Definition 3. Let $\eta = (a, b)$ be a p,q-QOFN. The score function (ϕ) of a p,q-QOFN can be mathematically stated as follows:

$$\phi = \frac{1 + a^p - b^q}{2}. \tag{2}$$

Definition 4. Let $\eta_j = (a_j, b_j)$ be any collection of p,q-QOFNs. The p,q-QOF weighted arithmetic (p,q-QOFWA) operator is given as

$$p,q - \text{QOFWA}(\eta_1, \eta_2, \cdots, \eta_n) = \left(\sqrt[p]{1 - \prod_{j=1}^{n}\left((1 - a_j)^p\right)^{w_j}}, \prod_{j=1}^{n}(b_j)^{w_j}\right), \tag{3}$$

where w_j is a weight vector such that $0 \le w_j \le 1$and $\sum_{j=1}^{n} w_j = 1$.

Definition 5. Let $\eta_j = (a_j, b_j)$ be any collection of p,q-QOFNs. The p,q-QOF Weighted Geometric (p,q-QOFWG) operator is obtained as

$$p,q - \text{QOFWG}(\eta_1, \eta_2, \cdots, \eta_n) = \left(\prod_{j=1}^{n}(a_j)^{w_j}, \sqrt[q]{1 - \prod_{j=1}^{n}\left((1 - b_j)^q\right)^{w_j}}\right), \tag{4}$$

where w_j is a weight vector such that $0 \le w_j \le 1$and $\sum_{j=1}^{n} w_j = 1$.

The p,q-QOFSs offer a higher degree of precision compared to other fuzzy models, such as Intuitionistic Fuzzy Sets (IFSs), Pythagorean Fuzzy Sets (PFSs), or even q-QOFSs, which can be further examined from the following aspects:

- **Boundary Crossing Uncertainty:** Unlike other fuzzy models, such as IFS and PFS, which are restricted by fixed linear or quadratic boundaries, p,q-QOFSs offer a mathematical approach for situations when experts are certain and, at the same time, risk-aware.
- **Decoupled Sensitivity (*p* vs. *q*):** Unlike q-QOFSs, p,q-QOFSs offer the advantage of using two different power values for membership and non-membership functions, which is a crucial feature of PCCs, as the cost of a false positive (membership) and the cost of a false negative (non-membership) are rarely equal in the context of pharmaceutical safety.
- **Granular Hesitancy Control:** p,q-QOFSs offer the advantage of using two power values, p and q, to stretch the fuzzy space so as to accommodate the extreme hesitancy of experts, such that the ranking is not overly sensitive.

These features represent not only a methodological improvement over existing fuzzy approaches, but also a conceptual shift in how expert judgment is modeled in high-stakes decision environments. The independent sensitivity of p and q allows the model to capture asymmetry in expert evaluations, which is particularly important in PCC technology assessment, where false positives and false negatives carry unequal consequences for patient safety and regulatory compliance. This alignment between the mathematical structure of p,q-QOFSs and the decision-making requirements of PCC resilience highlights a key distinction from existing fuzzy MCDM studies in supply chain management.

RAWEC method based on the p,q-QOFSs

Although the mathematical formulation of the p,q-QOF RAWEC method is presented in the following section, the overall procedure follows a clear three-stage logical structure:

The Aggregation Phase (Steps 1-4): The mathematical synthesis of various expert opinions to arrive at a single "group perspective" is conducted in this phase. The p,q-QOFWA operator is employed to ensure that there is no "skewed" outcome as well as to take account of seniority and levels of expertise.

The Normalization Phase (Steps 5-6): The various criteria (for instance, technical accuracy and investment costs) are standardized in this phase. This ensures a "fair comparison" between benefit-oriented and cost-oriented factors.

The Prioritization Phase (Steps 7-8): The final phase involves an evaluation of how far each of the AI strategies is from an "ideal" solution. The Q_i score is arrived at, providing an unambiguous ranking that represents the entire contribution to system resilience.

The RAWEC approach, as one of the recent MCDM models, efficiently streamlines decision-making by minimizing processes and evading intricate calculations (Puška et al., 2024). It includes four stages, with the initial two being universal to all methodologies. It integrates two procedures identified in alternative methodologies: (*i*) the formulation of a weighted normalized decision matrix and (*ii*) the computation of deviation from ideal and anti-ideal values. This procedure comprises the subsequent steps:

Step 1: In the p,q-QOF information-based MCDM process, let the sets of alternatives be defined as $\{A_1, A_2, \ldots, A_m\}$ and the criteria as $\{C_1, C_2, \ldots, C_n\}$. Define Exp as the set $\{Exp_1, Exp_2, \ldots, Exp_l\}$, representing l experts responsible for evaluating each alternative A_i according to the criterion C_j.

Step 2: Experts perform a linguistic assessment of the alternatives utilizing the linguistic scale. The assessments are subsequently converted into p,q-QOFNs based on the established scale. Let the $\Xi^l = \left[x_{ij}^l\right]_{m \times n}$, where $x_{ij}^l = \left(a_{ij}^l, b_{ij}^l\right)$, $l = 1,\ 2,\ \ldots, L$; $i = 1, 2, \ldots, m$; $j = 1, 2, \ldots n$. In this matrix, x_{ij}^l be defined by the experts, where x_{ij}^l represents the assessment value of alternative i, with regard to criterion j by expert l.

Each pair x_{ij}^{l} is assigned a value from the established p,q-QOFNs scale.

Step 3: The performance assessments of experts are presented as linguistic ratings, which are subsequently converted into p,q-QOFNs. Let $x_{ij}^{l} = \left(a_{ij}^{l}, b_{ij}^{l}\right)$ be a $p,\ q$-QOFN. Then, the procedure for deriving the numerical significance value of the lth expert is outlined as follows:

$$\phi_{ij}^{l} = \frac{1 + \left(a_{ij}^{l}\right)^{p} + \left(b_{ij}^{l}\right)^{q}}{2}. \tag{5}$$

Step 4: The Ξ^{l} matrices are aggregated by using the weights (w_j) of each expert using the p,q-QOFWA operator as in Eq. (3).

Step 5: A matrix of score measures for the criteria is created, as specified in Eq. (5), leading to the creation of the initial decision matrix.

$$\overline{\phi} = \begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} & \cdots & \phi_{1n} \\ \phi_{21} & \phi_{22} & \phi_{23} & \cdots & \phi_{2n} \\ \phi_{31} & \phi_{32} & \phi_{33} & \cdots & \phi_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \phi_{m1} & \phi_{m2} & \phi_{m3} & \cdots & \phi_{mn} \end{bmatrix}, \tag{6}$$

where $\phi_{ij} = \frac{1 + a_{ij}^{l\,p} + b_{ij}^{l\,q}}{2}$.

Step 6: Calculate the normalization of the initial decision matrix with double normalization through the application of Eqs. (7) and (8):

$$\vartheta_{ij} = \frac{x_{ij}}{\max(x_{ij})};\ \ \vartheta_{ij}' = \frac{\min(x_{ij})}{x_{ij}} \ \ \text{for}\ B, \tag{7}$$

$$\vartheta_{ij} = \frac{\min(x_{ij})}{x_{ij}};\ \ \vartheta_{ij}' = \frac{x_{ij}}{\max(x_{ij})} \ \ \text{for}\ C, \tag{8}$$

where$B$ and C denote benefit and cost criteria, respectively, while $\min(x_{ij})$ signifies the minimal value of alternatives based on specific criteria, and $\max(x_{ij})$ indicates the highest value of alternatives according to those criteria.

Step 7: Computing the deviation from the criterion weight. This stage integrates the weighting of the normalized decision matrix with the computation of the deviation from the criteria weights. This is achieved with the help of Eqs. (9) and (10):

$$\Bbbk_{ij} = \sum_{i=1}^{n} w_j\left(1 - \vartheta_{ij}\right), \tag{9}$$

$$\Bbbk_{ij}' = \sum_{i=1}^{n} w_j\left(1 - \vartheta_{ij}'\right), \tag{10}$$

where the criteria weight is denoted by w_j.

Step 8: In this final step, a comprehensive assessment score (Q_i) is calculated for each alternative. This score synthesizes the alternative's performance across all criteria by integrating its proximity to both the ideal and anti-ideal solutions, which were calculated as deviation values in Step 7. This is determined using Eq. (11):

$$Q_i = \frac{\Bbbk_{ij}' - \Bbbk_{ij}}{\Bbbk_{ij}' + \Bbbk_{ij}}. \tag{11}$$

Analysis and results

To demonstrate the application of the suggested MCDM model, a numerical analysis is performed to evaluate and enhance the resilience of the cold chain through AI-driven strategies.

Experts, criteria, and AI-driven strategies

Recognition of experts

The initial stage of the designed model, intended as a planning procedure, involved the formulation of a series of research questions as outlined in the introduction. Researchers subsequently established both a board of experts and a board of industry professionals. Both groups possess significant expertise and substantial practical understanding in the implementation of digital transformation and the improvement of resilience in cold chain operations. All experts are professionally active in Turkey, primarily within healthcare, logistics, and technology-intensive organizations. Although the board of experts, which comprises seven members, was established to examine the selection criteria and decision alternatives, a separate board of experts, consisting of three members, was created to analyze the weights of the selection criteria.

To effectively develop and implement AI-driven strategies for enhancing the resilience of the cold chain, assembling a multidisciplinary working group is essential. By assembling such a multidisciplinary team, we'll be well-equipped to perform our suggested model effectively, ensuring that our research is derived from real-world expertise and addresses the complexities of implementing AI-driven strategies in the cold chain. This group brings together a diverse array of expertise to address the complex challenges involved. Academic researchers and consultants provide theoretical insights and methodological rigor, ensuring the research is based on the latest scientific advancements. It is in this respect that industrial engineers add knowledge of the processes and how to optimize systems in cold chain environments. On the other hand, software engineers and AI experts develop and integrate these advanced technological solutions, adapting AI applications to cold chain logistics-specific needs. Supply chain managers and cold chain logistics experts provide practical insights regarding logistical challenges and requirements, ensuring that the proposed strategies are both feasible and effective in real-world settings. Meanwhile, AI and ML specialists focus on designing predictive models and algorithms to enhance decision-making and operational efficiency. Finally, IT integration experts ensure that new technologies are perfectly incorporated into existing systems and address any technical issues that arise. This multidisciplinary approach enables the working group to tackle technical, operational, and strategic challenges associated with implementing AI-driven solutions in CCM. This will lead to a much more robust and resilient supply chain.

The requirements were a minimum of 3 years of experience in PCC, requisite experience and knowledge in digital transformation operations, and experience as a manager or academician in cold chain operations and digital transformation. Based on our investigation, we found 25 applicants who completely fulfilled the criteria and designated these experts to join the expert board as associates. A total of 14 candidates responded favorably to our request and agreed to participate as expert committee members in the research. Table 1 includes comprehensive information about the chosen experts.

To be specific, experts were chosen on the basis of the following criteria: (i) having considerable professional or academic experience in areas pertinent to the study, such as pharmaceutical logistics, supply chain management, AI, or information systems; (ii) involvement in decision-making or technology implementation concerning supply chain operations; and (iii) possessing a minimum level of professional experience. Moreover, to ensure diversity of views, the expert panel consisted of professionals from different fields, including pharmaceutical supply chain management, industrial engineering, logistics management, AI/data analytics, IT systems, etc.

In Delphi-based studies, a panel of 10 to 15 experts is methodologically adequate to ensure the achievement of stable consensus while maintaining the depth of qualitative research. Our final panel consists of 14 experts, which is well above the minimum threshold of a panel of experts in high-level strategic decision-making in specific sectors, such as PCCs. Finally, it is worth noting that, in studies using experts, a certain level of domain bias is a possibility. To address this, the panel was composed of experts from different professional backgrounds pertaining to the subject matter, i.e., pharmaceutical supply chain management, along with experts in digital technology.

Table 1
Information regarding the experts in PCCs.

Expert	Profession	Experience	Expert	Profession	Experience	Expert	Profession	Experience
Exp1	Academician	12	Exp6	Academician	8	Exp 11	Industrial engineer	16
Exp2	Consultant	24	Exp7	Supply chain manager	26	Exp12	Supply chain manager	20
Exp3	IE	29	Exp8	Cold chain logistics expert	14	Exp13	AI and ML expert	3
Exp4	Software engineer	11	Exp9	AI and ML expert	17	Exp14	Cold chain logistics expert	12
Exp5	IT integration expert	15	Exp10	IT integration expert	32	-	-	-

Following the establishment of the decision-making committee, the first interview was conducted with the experts. During this interview, they were presented with a concise overview of the research scope and the anticipated procedures for the investigation. The researchers concluded the interview by explicitly acknowledging the perspectives and suggestions expressed by the experts during the initial session. Another important consensus reached at this symposium was that a decision-making model on PCCs, using the power of AI-driven technologies in terms of resilience, is urgently needed. According to the experts, while all criteria are important in the implementation process, there is a significant lack regarding how to render them truly useful. It was determined that once the right and proper criteria were established, the situation could be handled and resolved duly.

Determination of the relevant criteria

Building upon the initial orientation and evaluation session, various negotiation procedures started to be applied by the research team to set the desired criteria for evaluating AI-driven strategies in enhancing the resilience of PCCs. We employed the *p,q*-QOF Delphi method, a state-of-the-art extension of the traditional fuzzy Delphi approach, to further refine these criteria. This method utilizes the *p,q*-QOFSs that allow the experts to express their judgments more flexibly and precisely by capturing the membership and non-membership degree without compelling binary choices. With iterative questionnaires, experts evaluated the relevance of criteria in AI-driven strategies of cold chain resilience using linguistic terms that were converted to *p,q*-QOFNs. This countermanded the intrinsic uncertainty and hesitation in expert opinions due to the capturing of degrees of membership and non-membership without forcing a choice. The consolidated expert judgments thus allowed us to classify the criteria into critical, moderate, and uncritical groups based on consensus levels derived from the fuzzy assessments.

The criteria employed by this study were based on the resilience capabilities that have been established by supply chain resilience literature. To be specific, criteria such as real-time monitoring and data visibility support supply chain agility by enabling quick identification of temperature changes and disruptions. In addition, predictive analytics support supply chain adaptability by enabling quick identification of risks and disruptions. Similarly, alert systems and intelligent decision mechanisms support supply chain recovery by enabling quick responses to disruptions. In this regard, it is apparent that the criteria employed by this study have been able to operationalize key resilience concepts in relation to PCC management.

The progression from 47 candidate criteria to the final set followed a two-tier filtration process with distinct mathematical cutoffs:

- **Tier 1: The *p,q*-QOF Delphi Filtering Rule**

  Expert linguistic assessments were converted to *p,q*-QOFNs and aggregated using the *p,q*-QOFWA operator. To apply the cutoff, the defuzzified score was normalized to a 0-1 scale. Criteria with scores above a predefined consensus threshold were labeled "Critical" and passed automatically. Those below a secondary threshold were labeled "Uncritical" and discarded.

- **Tier 2: CVR Filtering for “Moderate” Criteria**

  Criteria falling in the “Moderate” middle ground underwent Lawshe’s CVR test to resolve ambiguity. For a panel of 14 experts, a minimum CVR of 0.51 was required for a criterion to be considered "Essential" and retained for the final model.

A panel of 14 experts in adequate disciplinary balance agreed to participate in this study. These experts, through iterative rounds of questionnaires, assessed the relevance of various criteria related to cold chain resilience using a *p,q*-QOFN linguistic scale as delineated in Table 2. After this collection, comments by experts are provided and presented in the Supplementary Material. First, experts' linguistic judgments were aggregated using the *p,q*-QOFWA operator. The defuzzification of the obtained aggregated vectors was done according to Eq. (2). We estimated the normalized importance value for each criterion by dividing each defuzzified value by the summation of all defuzzified values. In addition, each normalized importance value is divided by the maximum value in the index to standardize the scale. These fuzzy assessments were used to derive a consensus level, and based on this consensus, the criteria were categorized into three categories: critical, moderate, and uncritical. The criteria identified are compiled and presented in Table 3.

Furthermore, we gathered professional viewpoints regarding the significance of each criterion.

To further refine the moderate criteria identified through the *p,q*-QOF Delphi process, we conducted Lawshe's CVR test. To ensure relevance and essentiality, the CVR Validity Ratio is one of the most popular techniques for evaluating the content validity of suggested evaluation criteria. The technique is used to check whether a particular criterion is considered essential by a group of experts. This test quantitatively measures the content validity of each criterion by assessing the degree of expert agreement on its necessity. Experts rated each moderate criterion as “essential,” “useful but not essential,” or “not necessary”. The CVR was then calculated for each criterion to statistically determine its relevance. This step was crucial because it allowed us to objectively validate the essentiality of the moderate criteria, ensuring that only those with a high level of expert consensus were included in the final analysis. By ensuring that each retained criterion is both representative of the construct under study and considered essential by subject matter experts, the overall rigor and credibility of the analysis are strengthened. The empirical results, including the final set of nine selected criteria, are reported in Table 4, while the findings related to PCC resilience are presented in Table 5.

The finalized list of criteria consists of the most essential factors agreed upon by the experts to evaluate AI-driven strategies for enhancing cold chain resilience. These criteria are grouped into four distinct categories, i.e., *Technical, Business, Environmental*, and *Social* aspects, to comprehensively address all dimensions of CCM. This

Table 2
Linguistic scale for applying the *p,q*-QOF Delphi method.

Linguistic variable	$p,\ q$-QOFNs
Not important	(0.20, 0.98)
Slight important	(0.50, 0.96)
Moderately important	(0.45, 0.55)
Important	(0.81, 0.42)
Very important	(0.98, 0.15)

Table 3
Specified criteria as a result of the *p,q*-QOF Delphi process.

Code	Criteria	Group
NC1	Controlling temperature breakdown	Critical
NC4	Enhancement in cold chain monitoring and visibility	Critical
NC6	Ensuring last-mile delivery	Critical
NC9	Technology intensity	Critical
NC18	Predictive accuracy	Critical
NC22	Waste management and reduction	Critical
NC24	Multi-skilled workforce	Critical
NC25	Quality management system	Critical
NC28	Investment capacity	Critical
NC34	Cybersecurity	Critical
NC35	Energy consumption and efficiency	Critical
NC38	Firm agility	Critical
NC42	Improvement of operational efficiency and employee Scheduling	Critical
NC43	Occupational safety risk mitigation in cold chain facilities	Critical
NC46	Adopters' acceptance rate	Critical
NC2	Ensuring quality packaging material	Moderate
NC3	Monitoring the delivery process	Moderate
NC10	Observability	Moderate
NC13	Determine insufficient quantity of refrigerant	Moderate
NC14	Data reliability and sustainability	Moderate
NC19	Response time	Moderate
NC27	Public–private collaboration	Moderate
NC29	Firm size and structure	Moderate
NC47	Human capital	Moderate
NC5	Inadequate temperature monitoring and maintenance systems	Uncritical
NC7	Cold chain tracing	Uncritical
NC8	Lack of latest technology or optimal equipment	Uncritical
NC11	Trialability	Uncritical
NC12	Identifying worn or damaged coolbox	Uncritical
NC15	Information collection	Uncritical
NC16	Intelligent decision	Uncritical
NC17	Future prospects for the use of AI	Uncritical
NC20	Scalability	Uncritical
NC21	Integration complexity	Uncritical
NC23	Business certifications	Uncritical
NC26	Multi-sourcing	Uncritical
NC30	Business networks	Uncritical
NC31	Opinion leaders in the business	Uncritical
NC32	Internal and external economies of scale	Uncritical
NC33	Customs and norms in the business	Uncritical
NC36	Peer pressure	Uncritical
NC37	Expected returns	Uncritical
NC39	Cost-effectiveness	Uncritical
NC40	Regulatory compliance	Uncritical
NC41	Environmental concerns and actions	Uncritical
NC44	Inclusiveness	Uncritical
NC45	Workforce transformation	Uncritical

grouping ensures that our analysis considers operational efficiency, organizational capabilities, sustainability, and human factors. Definitions of the identified criteria are provided in Table 5 to facilitate a clear understanding of each aspect.

Determination of the AI-driven strategies

Experts and academics identified seven AI-driven technologies that can enhance resilience in PCCs. They unanimously agreed to assess the appropriateness of these solutions. Accordingly, we specified the seven AI-driven technologies below because they directly address the critical criteria we identified through our rigorous expert evaluation process. The key purpose of this research was to develop a model for long-term strategic decision-making by industry heads, such as supply chain executives and managers, who make decisions regarding the allocation of resources and investment in technology. The fundamental question we are trying to answer is not "Which specific algorithm should our data scientists use?" but "In which broad technological capabilities should we invest to make our cold chain more resilient?" It is here that we define our options at a strategic level, such as "Predictive Analytics and ML Models (A1)" or "AI-Powered Predictive Maintenance (A2)". Those are high-investment domains in infrastructure, talent, and process reengineering. Here is the list of seven AI-driven technologies identified by our experts:

1. Predictive Analytics and ML Models (A1)

Predictive analytics and ML models use historical and real-time data to predict potential cold chain failures, improving technical accuracy, temperature control, and data reliability. This strategy enhances business agility, reduces waste, supports environmental sustainability, and boosts adoption rates.

2. AI-Powered Predictive Maintenance (A2)

AI-powered predictive maintenance is an approach that uses sensor data to anticipate when maintenance will be required, therefore ensuring the operation of continuous work. It sets a higher standard for cost-effectiveness and environmental efficiency, as well as human capital, by reducing failure occurrences that would cause downtime or operational disruption.

3. AI-enabled Workforce Management (A3)

AI-enabled workforce management improves business effectiveness by creating optimized schedules and task assignments, utilizing multi-skilled workforces, increasing investment capacity, and ensuring environmental sustainability. It allows the development of a skilled workforce with greater adoption rates.

4. Real-Time Monitoring with IoT Integration (A4)

Real-time monitoring with IoT Integration uses IoT sensors and AI systems to enable a cold chain to offer real-time monitoring of temperature, humidity, and other critical parameters. It is expected to enhance the delivery process, monitoring, visibility, and quality of data. Supporting quality management by boosting operational efficiency and contributing to occupational safety with technology that maintains optimal conditions even for workers is better.

5. Anomaly Detection Systems (A5)

Anomaly detection systems operate based on patterns in operational data. They can point out abnormalities in the operations involved with cold chain operations, enhancing technical reliability to support quality management and guaranteeing smooth operation through addressing cybersecurity concerns.

6. Energy Optimization (A6)

Energy optimization utilizing AI optimizes energy consumption at the level of refrigeration units and warehouses, enabling smaller environmental impacts with higher intensiveness of technology applied, which is cost-effective in the solution. It also contributes to occupational safety and optimal working conditions.

7. Route Optimization Algorithms (A7)

Route optimization algorithms inspect the best routes of delivery, considering variables such as traffic flow, weather, and road conditions. The platform will further drive last-mile delivery efficiency and cold chain visibility, ensure business agility, reduce transit times and operational costs, decrease energy consumption and resulting emissions, and improve service levels and reliability.

The selection of these seven AI-driven strategies was guided by their direct alignment with the critical criteria identified through our rigorous expert evaluation process. Each strategy was chosen for its proven ability to enhance operational efficiency, ensure product integrity, and enable proactive decision-making within the PCC. By addressing the highest-weighted technical, business, environmental, and social criteria, these technologies offer practical and impactful solutions that mitigate risks, optimize resource utilization, and strengthen the overall resilience of CCM. Focusing on these key AI-driven strategies allows our research to provide a targeted and effective approach, ensuring both feasibility and significant positive outcomes for the industry.

Investigating the resilience of PCCs with the proposed model

The progressive use of the established model for the resilience of PCCs was executed in the case study as outlined below:

Table 4
Lawshe's content validity results for moderate criteria.

Code	Definition	Exp1	Exp2	Exp3	Exp4	Exp5
NC2	Ensuring quality packaging material	Essential	Useful but not essential	Useful but not essential	Essential	Not necessary
NC3	Monitoring the delivery process	Essential	Essential	Essential	Essential	Essential
NC10	Observability	Useful but not essential	Essential	Not necessary	Essential	Useful but not essential
NC13	Determine insufficient quantity of refrigerant	Essential	Essential	Essential	Not necessary	Not necessary
NC14	Data reliability and sustainability	Not necessary	Essential	Essential	Essential	Essential
NC19	Response time	Useful but not essential	Essential	Essential	Not necessary	Not necessary
NC27	Public–private collaboration	Essential	Essential	Essential	Essential	Essential
NC29	Firm size and structure	Useful but not essential	Essential	Essential	Essential	Essential
NC47	Human capital	Essential	Essential	Useful but not essential	Essential	Essential
Code	**Definition**	**Exp6**	**Exp7**	**Exp8**	**Exp9**	**Exp10**
NC2	Ensuring quality packaging material	Essential	Not necessary	Essential	Not necessary	Essential
NC3	Monitoring the delivery process	Essential	Essential	Essential	Essential	Essential
NC10	Observability	Essential	Useful but not essential	Essential	Not necessary	Essential
NC13	Determine insufficient quantity of refrigerant	Not necessary	Essential	Essential	Essential	Essential
NC14	Data reliability and sustainability	Essential	Essential	Essential	Not necessary	Essential
NC19	Response time	Essential	Not necessary	Essential	Useful but not essential	
NC27	Public–private collaboration	Essential	Essential	Essential	Essential	Essential
NC29	Firm size and structure	Essential	Essential	Essential	Useful but not essential	
NC47	Human capital	Essential	Essential	Useful but not essential	Essential	Essential
Code	**Definition**	**Exp11**	**Exp12**	**Exp13**	**Exp14**	**CVR**
NC2	Ensuring quality packaging material	Essential	Not necessary	Essential	Essential	0.1429
NC3	Monitoring the delivery process	Essential	Essential	Essential	Useful but not essential	0.8571
NC10	Observability	Useful but not essential	Essential	Essential	Useful but not essential	0
NC13	Determine insufficient quantity of refrigerant	Essential	Essential	Useful but not essential	Useful but not essential	0.2857
NC14	Data reliability and sustainability	Essential	Essential	Essential	Essential	0.7143
NC19	Response time	Essential	Essential	Not necessary	Essential	0
NC27	Public–private collaboration	Useful but not essential	Essential	Useful but not essential	Essential	0.7143
NC29	Firm size and structure	Essential	Essential	Useful but not essential	Essential	0.4286
NC47	Human capital	Essential	Useful but not essential	Essential	Essential	0.5714

Assessing the weight assigned to each expert

A mapping is proposed to build a correlation between the number of years of experience and the corresponding linguistic characteristics in order to assess the expertise of scholars in p,q-QOFNs. An assessment of the experience levels of scholars is carried out by taking into account the linguistic factors specified in Table 6. Considering the specified range of linguistic variables, which extends from "Extremely Low (EL)" to "Extremely High (EH)", we define intervals of experience years that correspond to each linguistic variable.

EL is defined as having 0-3 years of experience, VL as having 4-7 years, L as having 8-11 years, UA as having 12-15 years, A as having 16-19 years, AA as having 20-23 years, H as having 24-27 years, VH as having 28-31 years, and EH as having 32 years and older. The experience of experts was determined using linguistic variables presented in Table 6. At first, based on the defined intervals, we assigned each expert a linguistic variable that corresponds to their years of experience. After that, we derived the crisp values of experts' importance through the score function indicated in Eq. (2). Finally, we implemented the linear sum normalization to obtain the experts' importance levels, as shown in Table 7.

The results in Table 7 represent how the proposed methodology systematically translates the qualitative 'experience' attribute into quantitative weights that influence the results of the study. A comparison between Exp10 and Exp13 best describes this process:

- **Linguistic and Fuzzy Evaluation:** Based on their respective experiences of 32 and 3 years, Exp10 assigned the linguistic variable “Extremely High Importance (EH),” while Exp13 assigned “Extremely Low Importance (EL)”. As provided in Table 6, they are the p,q-QOFNs of (0.99, 0.11) for Exp10 and (0.11, 0.99) for Exp13. In Exp10, the membership degree (μ); i.e., importance, is very high (0.99), and the non-membership degree (ν) is very low (0.11). It is the opposite for Exp13.
- **Conversion to Crisp Values:** These p,q-QOFNs are converted to single “crisp values” through the score function from our method. The score function is designed to provide a high value to p,q-QOFNs with high membership and low non-membership. So, Exp10′s p,q-QOFN of (0.99, 0.11) yields a high crisp score of 0.9796, while Exp13′s (0.11, 0.99) yields a very low crisp score of 0.0149.
- **Impact on Final Weights:** These crisp scores are normalized to derive the final expert weights. Due to the large difference in their crisp scores, Exp10 receives the maximum weight in the group (0.1502, or 15.02%), while Exp13 receives the minimum (0.0023, or 0.23%). This directly influences the subsequent stages of the analysis; for example, when summing up the criteria scores in Table 8,

Table 5
Definition of identified criteria.

Criterion	Definition	References
Technical Aspects		
Controlling temperature breakdown (C1)	Ensuring that temperature-sensitive products remain within the required temperature range throughout the cold chain to prevent spoilage or degradation. It is critical in cold chain logistics, as it ensures the integrity and quality of perishable goods, preventing spoilage and maintaining efficacy, especially for food and pharmaceuticals.	(Casula et al., 2024; Skawińska and Zalewski, 2022)
Monitoring the delivery process (C2)	Continuously tracking the transportation and handling of goods to ensure timely delivery and adherence to temperature and quality standards.	(Niu et al., 2023)
Enhancement in cold chain monitoring and visibility (C3)	Improving real-time tracking and transparency across all stages of the cold chain to quickly identify and address potential issues.	(Ruiz García, 2008)
Ensuring last-mile delivery (C4)	Guaranteeing the efficiency and reliability of the final segment of the delivery process, where products reach their end destination.	(Wanganoo et al., 2021)
Technology intensity (C5)	Level of advanced technologies, such as AI and automation, are integrated into cold chain operations to improve efficiency and accuracy.	(Lam et al., 2023)
Data reliability and sustainability (C6)	Maintaining high-quality, accurate data over time to support decision-making and ensuring data practices are secure and environmentally sustainable.	(Chen et al., 2022; Raj et al., 2024)
Predictive accuracy (C7)	Capability of AI systems to accurately forecast future events or conditions within the cold chain, enables proactive interventions.	(Lorenc et al., 2021)
Business Aspects		
Multi-skilled workforce (C8)	Employees possess a diverse set of skills, enabling them to perform various tasks and adapt to different roles within the cold chain operations.	(Ali et al., 2021)
Quality management system (C9)	A formalized system that documents processes, procedures, and responsibilities for achieving quality policies and objectives to enhance customer satisfaction.	(Thakur et al., 2024)
Public–private collaboration (C10)	Partnerships between government entities and private sector companies to utilize resources and expertise for improving cold chain infrastructure and services.	(Ali et al., 2017)

Table 5 (*continued*)

Criterion	Definition	References
Technical Aspects		
Investment capacity (C11)	Financial ability of a firm to invest in new technologies, infrastructure, and training to enhance cold chain operations.	(Srinivasan and Yadav, 2024)
Firm agility (C12)	Ability of a company to quickly adapt to market changes or disruptions, maintaining operational effectiveness in the cold chain.	Expert opinion
Environmental Aspects		
Improvement of operational efficiency and employee scheduling (C13)	Enhancing processes and optimizing staff schedules to reduce resource waste, increase productivity, and minimize the environmental footprint.	(Lorenc et al., 2021)
Energy consumption and efficiency (C14)	Reducing the amount of energy used in cold chain operations and improving energy efficiency to lower costs and environmental impact.	(Shaharudin and Fernando, 2024)
Waste management and reduction (C15)	Implementing strategies to minimize waste generation and promote recycling or proper disposal within the cold chain processes.	(Davoudi et al., 2024; Edwin et al., 2022)
Social Aspects		
Occupational safety risk mitigation in cold chain facilities (C16)	Implementing measures to protect workers from health and safety risks associated with cold chain environments, such as extreme temperatures and equipment hazards.	Expert opinion
Adopters' acceptance rate (C17)	The degree to which employees and stakeholders are willing to embrace and effectively use new technologies or processes introduced in the cold chain.	Expert opinion
Human capital (C18)	The collective skills, knowledge, and experience of the workforce contribute to the organization's performance and innovation capacity.	Expert opinion
Cybersecurity (C19)	Protecting digital systems and sensitive data within the cold chain from cyber threats to ensure operational integrity and confidentiality.	(Garcia-Perez et al., 2023)

Exp10′s judgments will contribute approximately 65 times more influence on the outcome than Exp13′s judgments.

Establishing the criteria weights

Each expert evaluated the primary and secondary criteria using linguistic factors, as specified in Table 6. Table 8 presents the assessments performed by experts regarding the primary criteria. Subsequently, we transformed these evaluations into *p,q*-QOFNs. We obtained the aggregated assessment matrix using the *p,q*-QOFWA operator, as detailed in Eq. (3). Score values were computed, and the consolidated evaluation matrix for the primary criteria was obtained in crisp form (Table 9). The

Table 6
Linguistic aspects applied to the assessment of experts and criteria.

Linguistic variable	*p,q*-QOFNs
Extremely high importance (EH)	(0.99, 0.11)
Very high importance (VH)	(0.88, 0.22)
High importance (H)	(0.77, 0.33)
Above average importance (AA)	(0.66, 0.44)
Average importance (A)	(0.55, 0.55)
Under average importance (UA)	(0.44, 0.66)
Low importance (L)	(0.33, 0.77)
Very low importance (VL)	(0.22, 0.88)
Extremely low importance (EL)	(0.11, 0.99)

Table 7
Linguistic importance evaluations of scholars based on their experience.

Experts	Evaluation	*p,q*-QOFNs	Crisp values	Normalize values
Exp1	UA	(0.44, 0.66)	0.3750	0.0575
Exp2	H	(0.77, 0.33)	0.6578	0.1009
Exp3	VH	(0.88, 0.22)	0.7945	0.1219
Exp4	L	(0.33, 0.77)	0.2777	0.0426
Exp5	UA	(0.44, 0.66)	0.3750	0.0575
Exp6	VL	(0.22, 0.88)	0.1604	0.0246
Exp7	H	(0.77, 0.33)	0.6578	0.1009
Exp8	UA	(0.44, 0.66)	0.3750	0.0575
Exp9	A	(0.55, 0.55)	0.4626	0.0709
Exp10	EH	(0.99, 0.11)	0.9796	0.1502
Exp11	A	(0.55, 0.55)	0.4626	0.0709
Exp12	AA	(0.66, 0.44)	0.5523	0.0847
Exp13	EL	(0.11, 0.99)	0.0149	0.0023
Exp14	UA	(0.44, 0.66)	0.3750	0.0575

Table 8
Linguistic evaluations for primary criteria.

Experts	Technical	Business	Environmental	Social aspects
Exp1	VH	AA	L	VL
Exp2	VH	H	A	AA
Exp3	EH	A	AA	A
Exp4	VH	H	VL	L
Exp5	VH	AA	AA	A
Exp6	EH	VH	A	UA
Exp7	VH	UA	A	L
Exp8	H	A	L	AA
Exp9	EH	L	A	A
Exp10	H	VH	H	AA
Exp11	EH	VH	A	H
Exp12	VH	VH	AA	A
Exp13	EH	A	UA	L
Exp14	EH	VH	EL	A

Table 9
Aggregated evaluations and weights of primary criteria.

	Technical	Business	Environmental	Social aspects
***p,q*-QOFNs**	(0.9465, 0.1880)	(0.7805, 0.3642)	(0.6179, 0.5270)	(0.5973, 0.5367)
Score values	0.8980	0.6614	0.4997	0.4864
Final values	0.3528	0.2598	0.1963	0.1911

primary criterion is technical aspects, assigned a weight of 35.28%, followed by business aspects at 25.98%, environmental aspects at 19.63%, and social aspects at 19.11%.

In order to ascertain their local weights, comparable evaluations and calculations are implemented for secondary criteria. The global weights of each secondary criterion are determined by multiplying the local weights of the secondary criteria by the weight of the primary criteria. The results are presented in Table 10.

Evaluation of AI-driven strategies utilizing the RAWEC method

Upon establishing the significant levels of the criteria, we employ the *p,q*-QOF RAWEC approach to assess AI-driven technologies. The execution of the *p,q*-QOF RAWEC method adheres to the procedures outlined in Section 3.2. Experts assessed the technologies according to each secondary criterion with the help of Table 11 (see Table S1). Next, the assessments are converted into *p,q*-QOFNs that are given in Table S2. The aggregated decision matrix is computed using Eq. (3), and the outcomes are outlined in Table S3. The preliminary decision matrix for the criterion is obtained by acquiring the scores of technologies via Eq. (2). Table S4 displays the scores of technologies and is considered a decision matrix. Eqs. (7) and (8) are then applied to normalize the elements of the initial decision matrix. The deviation of the criteria weights is computed using Eqs. (9) and (10), employing the previously determined weight of the *p,q*-QOFWA operator. The alternatives value (Q_i) are ultimately derived via Eq. (11), and the conclusive ranking scores of the technologies are provided in Table 12.

Examination of the robustness

The solutions to the issue obtained with the suggested methodology will be compared in terms of consistency, validity, and robustness by altering the expert weight values and the criteria weight values, conducting tests for the existence of the rank reversal problem, and comparing the solutions obtained with different MCDM methods. The performance of the proposed model is evaluated and validated through a comprehensive sensitivity analysis that is conducted in four stages.

The initial stage examines the effects of adjusting the weights of the experts on the classification results. The weight of each expert is progressively reduced from 10% in each scenario until it reaches zero. The weights of the remaining experts are adjusted to ensure that the sum of their weights is equal to 1. Accordingly, an analysis is conducted to study the impacts of alterations in the weights of experts on the ranking results. The ranking performance for all alternatives in all scenarios remains unaltered, as illustrated in Fig. 2. Consequently, the aggregate ranking results are not affected by changes in the weights of the experts. It demonstrates that the proposed model is entirely stable and consistently alters the relative significance degrees of the experts.

The primary criteria weights were adjusted in the second phase of the validation test to investigate how these adjustments affected the ranking outcomes. Forty scenarios were generated by lowering the weight of every criterion by ten percent in every scenario in order to do this, as

Table 10
Final weights of primary and secondary criteria.

Primary Criteria	Weights	Secondary criteria	Weights	Global Weights
Technical	0.3528	**C1**	**0.2327**	**0.0821**
		C2	0.1835	0.0647
		C3	0.1820	0.0642
		C4	0.1370	0.0483
		C5	0.1144	0.0404
		C6	0.0785	0.0277
		C7	0.0719	0.0254
Business	0.2598	C8	0.1501	0.0390
		C9	**0.1912**	**0.0497**
		C10	0.1023	0.0266
		C11	0.2976	0.0773
		C12	0.2587	0.0672
Environmental	0.1963	C13	0.3088	0.0606
		C14	**0.5012**	**0.0984**
		C15	0.1900	0.0373
Social	0.1911	C16	0.2053	0.0392
		C17	0.1501	0.0287
		C18	0.2936	0.0561
		C19	**0.3511**	**0.0671**

Table 11
Linguistic scale applied in the assessment of criteria.

Linguistic variable	$p,\ q$-QOFNs
Very Bad (VB)	(0.20, 0.98)
Bad (B)	(0.50, 0.96)
Medium (M)	(0.45, 0.55)
Good (G)	(0.81, 0.42)
Very Good (VG)	(0.98, 0.15)

Table 12
Results of p,q-QOF RAWEC method.

Technologies	ν_{ij}	$\nu_{ij}^{'}$	Q_i	Rank
A1	0.185	0.240	0.128	1
A2	0.193	0.224	0.072	3
A3	0.258	0.171	-0.204	7
A4	0.192	0.225	0.080	2
A5	0.201	0.219	0.042	4
A6	0.213	0.200	-0.032	6
A7	0.205	0.220	0.035	5

illustrated in Fig. 3.

During the third phase of the sensitivity analysis, the suggested p,q-QOFS-based approach produced findings that were compared to ranking outcomes obtained by several p,q-QOFS-based MCDM techniques. Various p,q-QOFS-based MCDM techniques were used in this regard, including p,q-QOFS-based Additive Ratio Assessment (p,q-ARAS), p,q-QOFS-based Multi-Attributive Border Approximation area Comparison (p,q-MABAC), p,q-QOFS-based Technique for Order Performance by Similarity to Ideal Solution (p,q-TOPSIS), and p,q-QOFS-based Weighted Aggregates Sum Product Assessment (p,q-WASPAS). Fig. 4 shows the ranking results of these p,q-QOFS-based MCDM techniques, as well as the proposed p,q-QOFS-based methodology.

Discussion

This study demonstrates that the integration of AI-driven technologies within a p,q-QOF-based MCDM framework provides a systematic means to prioritize resilience-enhancing strategies for PCCs. In this study, we sought to investigate the vital role of AI-based strategies in reinforcing the resilience of the PCC management. By using advanced fuzzy MCDM techniques, such as the p,q-QOF fuzzy Delphi process and

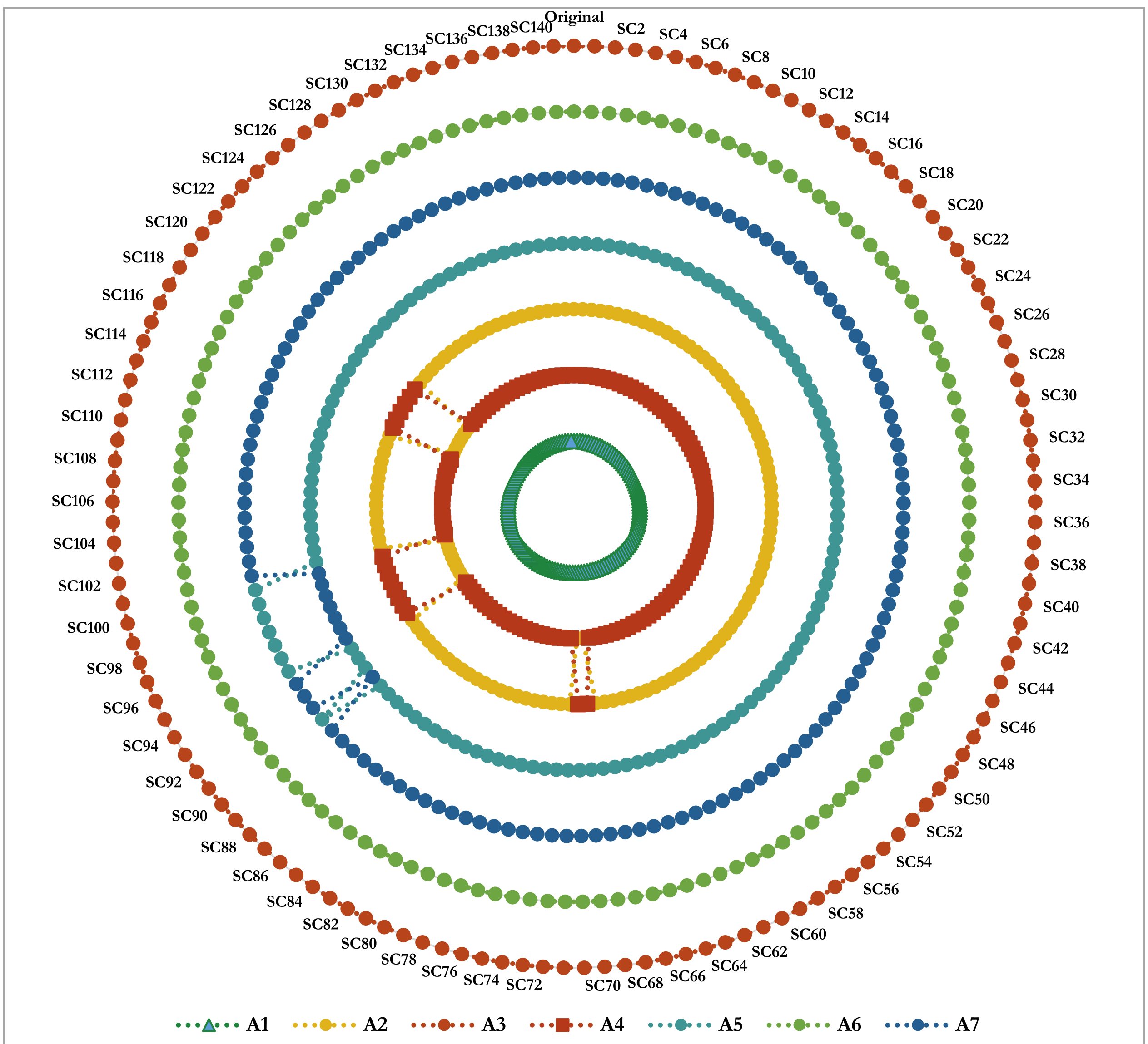


Fig. 2. Updated classification results of alternatives regarding the modification of the weights of experts for all scenarios.

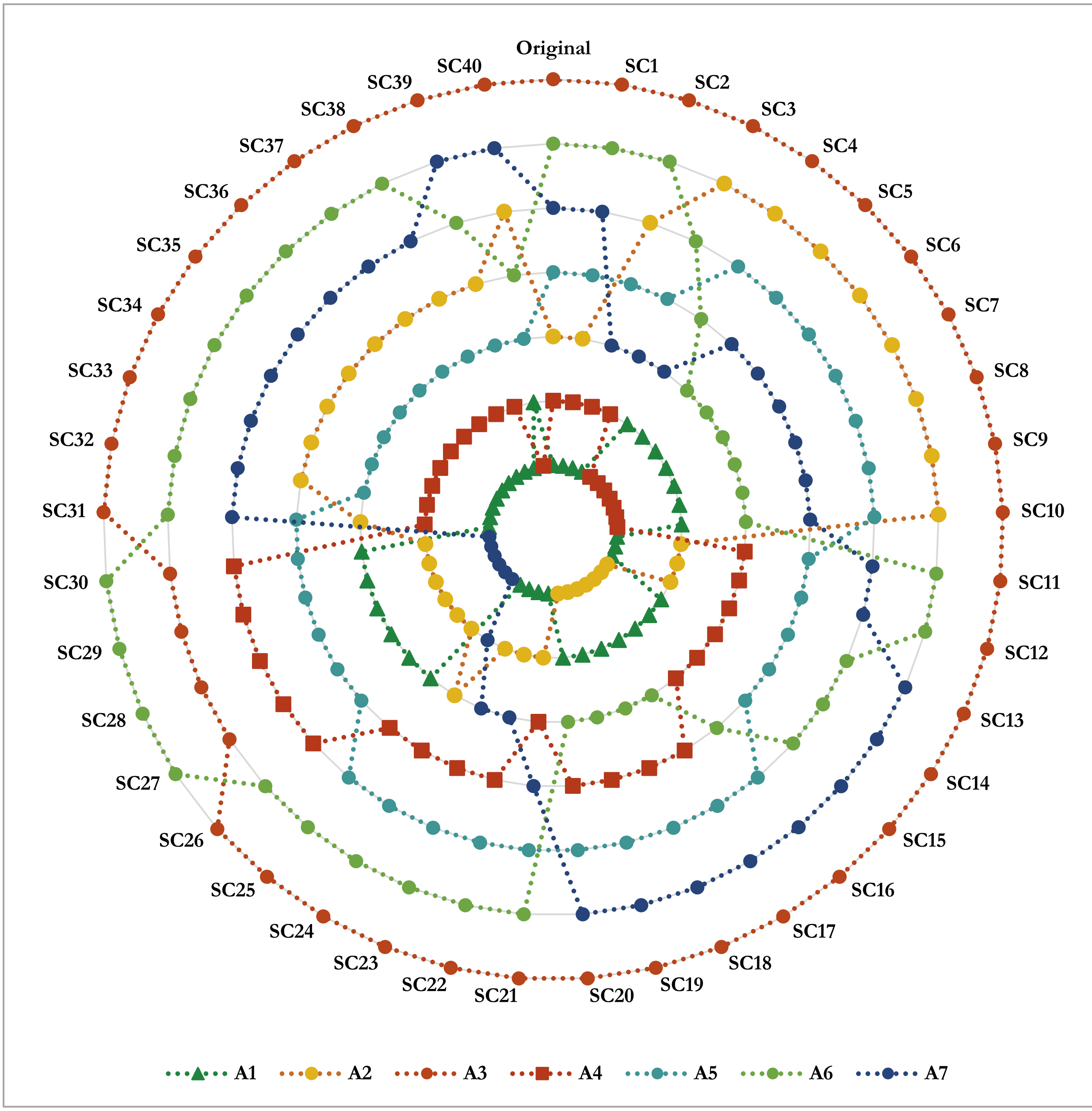


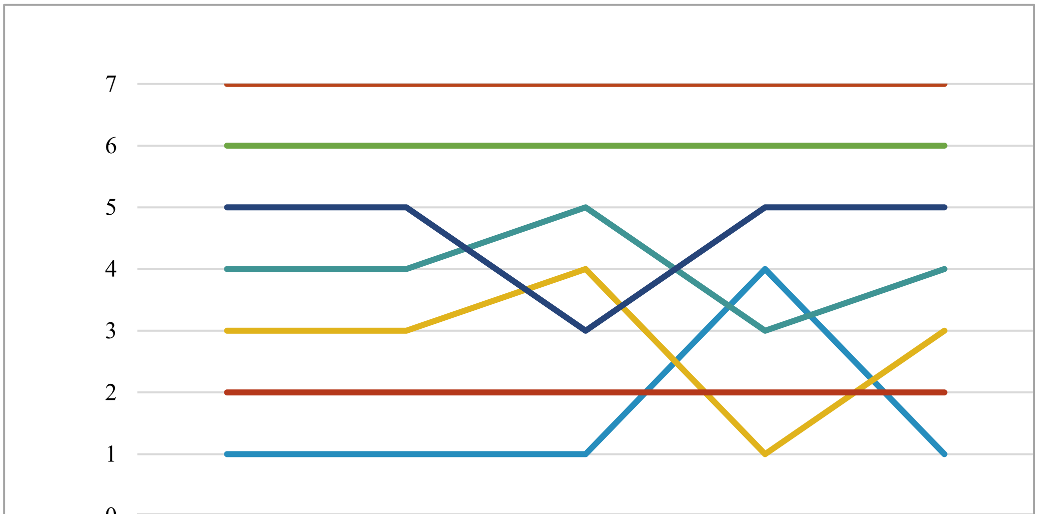
Fig. 3. Updated classification results of alternatives regarding the modification of the weights of primary criteria for all scenarios.

Lawshe's content validity test, we were able to identify and prioritize the most vital criteria affecting the resilience of the cold chain based on the context. The expert group, consisting of 14 experts from various fields, enabled us to gain deeper knowledge of the various criteria, which we classified under various aspects such as technical, business, environmental, and social aspects.

The results clearly show that the use of predictive analytics, real-time IoT-enabled monitoring, and AI-based anomaly detection are the most influential factors, thus confirming the vital role they play in ensuring the continuity of operations and mitigating any potential risk (Lorenc et al., 2021; Lau et al., 2021; Golan et al., 2021). This finding is in line with the emerging literature on the need for digital traceability, risk prediction, and predictive maintenance as enablers of robust and adaptive cold chain systems (Bamakan et al., 2021; Maier et al., 2024; Thakur et al., 2024).

Significantly, it is observed from the results that it is not just the developments in technology that are required to be made for sustainable resilience, but also human capital development. This aligns with the broader sustainable technology literature, which increasingly recognizes that social and organizational preparedness can be just as important as technical performance in shaping technology-enabled resilience (Raman et al., 2025). The high weightage of workforce criteria also implies that it is essential to have multi-skilled resources to effectively handle advanced monitoring systems, interpret data insights, and respond to any irregularities detected in the data, which is in line with earlier studies that emphasize that for any digital transformation to be implemented, organizational preparedness is also required to adopt technology (Ali et al., 2021; Bellantuono et al., 2021).

From a managerial point of view, the current study presents a roadmap for decision support for managers. First, resource allocation should focus on predictive modeling and real-time monitoring technologies, as they provide the greatest improvement in resiliency for the investment (Jackson et al., 2025; Wang et al., 2016). Second, organizations should establish ongoing workforce training and experiential learning opportunities to facilitate swift action on early warning signs provided by AI systems (Ali et al., 2017; Wong et al., 2023). Thirdly, the



	p, q-RAWEC	p, q-ARAS	SF-MABAC	p, q-TOPSIS	SF-WASPAS
A1	1	1	1	4	1
A2	3	3	4	1	3
A3	7	7	7	7	7
A4	2	2	2	2	2
A5	4	4	5	3	4
A6	6	6	6	6	6
A7	5	5	3	5	5

Fig. 4. Comparison of the findings of several *p,q*-QOFS approaches with the suggested model.

aspect of cybersecurity and data integrity has to be factored in as part of the resilience planning process since the increased digitalization will inevitably increase the attack surface (Garcia-Perez et al., 2023; Khan & Ali, 2023). Integrating data sharing protocols, traceability solutions, and real-time alert systems will improve system reliability (Qian et al., 2022; Tsang et al., 2018). In terms of strategy, the study findings reinforce the significance of adopting a portfolio approach to the implementation of AI solutions while factoring in the feasibility of the solutions from a business perspective as well as the environmental and social dimensions (Zhang & Mohammad, 2024; Khalid et al., 2024). Aligning the investment in the digital solutions with the weighted criteria developed in the study will help organizations improve their resilience while avoiding cost escalations that could negatively impact competitiveness. To enhance the practical relevance of these findings, the following decision-oriented strategic roadmap is proposed:

- **Scenario A: Resource-Constrained Environments**

Organizations operating in such environments should prioritize Predictive Analytics and ML Models (A1), as these offer the highest potential resilience returns relative to investment. At the same time, investments in Human Capital (C18) and a Multi-skilled Workforce (C8) are essential to fully leverage the benefits of AI-driven insights without requiring immediate or extensive automation.

- **Scenario B: High Regulation/Regulatory Compliance**

In highly regulated pharmaceutical environments, where there is zero tolerance for temperature deviations, organizations should prioritize Real-Time Monitoring with IoT (A1) and Anomaly Detection (A5). These technologies provide a strong "absorptive foundation" for maintaining compliance and ensuring product integrity. In addition, Cybersecurity (C19) must be incorporated into resilience planning to safeguard data integrity and system reliability within increasingly digitalized operations.

- **Scenario C: Contextual Adaptation**

Managers are encouraged to use this framework as a flexible and robust template for adaptation, rather than as a rigid measurement tool. While Predictive Analytics (A1) serves as a foundational priority, the effective deployment of AI in unstable environments is likely to depend on capabilities such as Route Optimization (A7) and Agile Workforce Management (A3), which help organizations respond to and mitigate systemic disruptions.

Furthermore, the prioritization findings can be used as guidelines for organizations seeking to improve the resilience of the PCC through the use of digital technologies. For example, in cases where the organization operates under limited financial constraints, the initial investment in the technologies can be in technologies such as AI-driven analytics systems, which are known for having strong prediction and monitoring abilities. This would be instrumental in improving the early detection of temperature changes and operational disruptions. For example, in cases where the organization operates in the pharmaceutical industry and has high regulatory needs, the use of AI-driven systems would be instrumental in improving the visibility of the cold chain data to improve regulatory compliance. The high position of the technologies in the prioritization results is highly linked to the technologies' abilities to effectively automate and improve the pillars of supply chain resilience:

1. **Anticipation (Proactive Capacity):** The main drivers that influence this capacity construct are "Predictive Analytics and ML Models", and "Anomaly Detection". These enable the PCC to "foresee" the disruptions even before they occur, which is critical since medical integrity can never be "recovered" once the temperature violations occur.
2. **Absorption (Robustness):** "AI-Powered Predictive Maintenance" and "Real-Time Monitoring with IoT" can be the facilitators that can

improve this capacity construct. This way, the supply chain can "absorb" the technical issues that occur while keeping the products at optimal temperature stability.

3. **Adaptation (Agility):** "Route Optimization Algorithms" and "AI-Powered Workforce Management" can be the facilitators that can improve this capacity construct, which is required to adapt to the crisis situation that has arisen.

It must be noted that technologies such as *Real-Time Monitoring with IoT, Anomaly Detection*, and *Predictive Analytics* also support the recovery capability (restoration capacity) of the PCC. These technologies enable the rapid identification of disruptions, temperature deviations, or operational failures, allowing managers to take corrective actions quickly. By providing timely alerts and data-driven insights, these systems facilitate faster response and restoration of normal supply chain operations while minimizing product loss and ensuring compliance with strict pharmaceutical regulations.

The findings of this study also offer a number of theoretical implications since it shows how fuzzy MCDM can be used in conjunction with AI applications in PCCs. The study provides a hierarchical approach that can be used as a guide for further research on digital supply chain resilience. The findings of this study, in terms of policy, illustrate how government support can be used to accelerate the integration of these technologies for better health outcomes. In terms of management, this work provides a guide for how resources can be allocated for these technologies that can improve supply chain resilience in a firm. The practical value of the framework lies less in its mathematical complexity and more in the quality of the decisions it supports. A lower-order fuzzy model applied to the same dataset would risk conflating technologies with fundamentally different risk profiles, for example, by treating a technically robust but organizationally immature solution as equivalent to one that is both technically sound and operationally ready. The *p,q*-QOF structure helps avoid this issue by enabling more differentiated and realistic prioritization. Similarly, the Delphi-CVR validation process ensures that rankings are based only on criteria that have achieved clear expert consensus regarding their essentiality. This strengthens practitioner confidence that the framework captures factors that are truly critical for pharmaceutical cold chain resilience.

To ensure maximal practical application of the prioritization results, a tri-step implementation strategy is proposed. The first step involves assessing organizational readiness in terms of digital infrastructure maturity, compliance with regulation, and workforce readiness before undertaking any technological investments to ensure that the deployment context is compatible with organizational capacities. At the second step, the decision about which technologies to use should be based on framework ratings together with organizational readiness levels: organizations facing resource constraints should focus on Predictive Analytics (A1) and Real-Time IoT Monitoring (A2), offering the highest returns for resilience across all four dimensions; organizations operating within highly regulated frameworks should also prioritize AI-Driven Anomaly Detection (A4); and organizations at early stages of digital transformation should undertake technology deployment in a phased manner starting with monitoring, followed by routing optimization and then workforce management. At the final step, post-deployment performance should be assessed based on the four resilience dimensions, while the framework may be reapplied to the new organizational realities, regulations, or technologies available. Both cybersecurity measures and data integration infrastructure should be viewed as enabling elements of all steps.

All in all, the study draws attention to enduring challenges such as interoperability among heterogeneous systems, scalability constraints in resource-limited contexts, and the need for stronger cross-sector collaboration (Peters & Sayin, 2023; Ochieng et al., 2024; Jafarian et al., 2025). Addressing these barriers will be essential to translate the proposed strategies into measurable and sustained performance gains across global PCC networks.

Conclusions and outlook

In this study, we sought to examine the vital function of AI-driven strategies in reinforcing the resiliency of PCC management. Using advanced fuzzy MCDM tools, such as the *p,q*-QOF fuzzy Delphi approach and Lawshe's content validity test, we were able to comprehensively identify and prioritize the most vital criteria that impact cold chain resiliency. Through the insights provided by our group of experts from different but relevant fields of study, we were able to classify these criteria under different technical, business, environmental, and social criteria groups. Through this analysis, we were able to pinpoint different AI-driven tools that are most aligned with the highest-ranked criteria. These tools include predictive analytics and ML tools, which were identified as having the highest weightage, followed by real-time monitoring with IoT integration, AI-driven predictive maintenance tools, anomaly detection tools, route optimization tools, and finally, energy optimization tools. These tools are all aligned with different critical technical aspects of cold chain resiliency, such as addressing temperature breakdown issues, improving monitoring and visibility, and resolving data reliability and accuracy issues (Lorenc et al., 2021; Lau et al., 2021). The results obtained in this research carry important theoretical implications for contributing to academic discussions by bridging the gap in fuzzy MCDM techniques and AI applications in CCM. The research offers a new approach in terms of a hierarchical framework for potential applications in subsequent research studies. The integration of fuzzy MCDM techniques with AI in CCM can be seen in a broader discussion on digital supply chain resilience and decision intelligence (Ivanov et al., 2019). In terms of policy implications, this research highlights the importance of government support in adopting impactful AI-based technologies, indicating a potential acceleration in the integration of technology with favorable policies for public health benefits. In terms of managerial implications, this research provides a guide for businesses on how to effectively channel resources into impactful technologies for enhancing resilience. Workforce development is critical for potential success (Ali et al., 2021; Wong et al., 2023).

The study contributes to the existing body of knowledge by closing the gap between theoretical frameworks and their applications in cold chain resilience. It highlights the significance of targeted AI technologies and offers a holistic approach by taking into account different facets of CCM. Further studies may be built upon this study by exploring the challenges of implementing these AI technologies and their long-term effects on SC performance (Tukamuhabwa et al., 2015). Moreover, exploring other industries that use cold chain logistics may also be beneficial in validating and enhancing the results of this study. To conclude, by incorporating AI technologies with the aid of a thorough evaluation of key criteria, it is possible to create a promising avenue to improve the resilience of PCCs to a considerable extent. By embracing these key technologies and addressing key critical issues, organizations may be able to improve their operational efficiency, ensure product integrity, and create a sustainable competitive advantage in a global market that is becoming increasingly complicated and challenging.

Although the proposed framework for enhancing PCC resilience using AI-driven strategies addresses key requirements and provides a practical managerial tool, it has several limitations that future research could examine. The focus on a limited set of AI-driven technologies excludes emerging tools such as generative AI and blockchain-enhanced systems (Ochieng et al., 2024; Bamakan et al., 2021), while the use of the *p,q*-QOF system adds complexity and needs to be compared to alternative fuzzy systems. Furthermore, the framework's other weaknesses are based on the expert panel's judgements. Even though the use of the Delphi method and Lawshe's CVR were aimed at generating a comprehensive consensus and eliminating bias, the outcome is naturally subject to the experience and viewpoint of the 14 experts involved. Moreover, experts from diverse geographical or professional backgrounds may propose a completely distinct set of prioritized criteria and technologies. The expert-based process may limit the generalizability of our specific outcomes. Our expert panel's diverse range of professional tasks may converge in a common region or economy, which implicitly shapes their perspectives. We acknowledge that practical challenges such as scalability, data integration, and generalizability to other industries or less technologically advanced areas are crucial considerations. Therefore, players in this industry within different settings of operations should consider this framework as a robust template for adaptation and experimentation based on advice from local experts rather than a universally absolute scale.

Another limitation concerns the number of experts involved in the Delphi study. Although the participation of 14 experts satisfies the minimum threshold recommended by Lawshe's formula for achieving statistically significant CVR values in Delphi-based MCDM studies, the sample size remains relatively limited for establishing universally generalizable priorities regarding AI solutions for pharmaceutical cold chains. A smaller expert panel may increase the likelihood that individual perspectives, professional experiences, or contextual biases influence the consensus-building and criteria-ranking processes. This concern is particularly relevant given the substantial differences in regulatory environments, organizational structures, technological maturity, and operational practices across pharmaceutical logistics systems worldwide. Nevertheless, the primary objective of this study was not to generate statistically representative findings or develop a predictive prioritization model. Rather, the study aimed to propose a strategic decision-support framework capable of assisting practitioners and policymakers in evaluating and prioritizing AI technologies within pharmaceutical cold chain contexts. Despite this intended scope, the limited number of experts should still be acknowledged as a constraint when interpreting the broader applicability and universality of the findings.

To overcome these limitations, future research could expand the range of AI-driven technologies and include methodological validation studies in which the effectiveness of the suggested *p,q*-QOF decision framework is compared with other approaches to modeling uncertainty in decision systems, such as intuitionistic fuzzy, q-rung orthopair fuzzy, or probabilistic decision systems. Such studies could help us better understand how effective this suggested framework is. Secondly, we suggest some possibilities for future studies that could include empirical research into the impact of AI technology adoption on the resilience of PCCs. By comparing how well these cold chains operate before and after adopting AI technologies, future studies could provide evidence of the improvements in resilience that our suggested framework found. Finally, the proposed *p,q*-QOF model can be applied in comparative studies across different industry sectors, such as food cold chains and biomedical logistics, where temperature-sensitive supply chains play a critical role.

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CRediT authorship contribution statement

Ahmet Çalık: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abdulaziz Alshalfan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erfan Babaee Tirkolaee:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

Supplementary materials

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